

A POLYNOMIAL-SCALE GAP BETWEEN THE CHROMATIC AND COCHROMATIC NUMBERS OF A RANDOM GRAPH

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ABSTRACT. For a graph G , let $\chi(G)$ be its chromatic number and let $\zeta(G)$ be the least number of parts in a vertex partition in which every part induces either an empty graph or a complete graph. We prove that, for $G_n \sim G(n, 1/2)$,

$$\mathbb{P}\left(\chi(G_n) - \zeta(G_n) \geq \frac{(\ln 2)^2}{32} \ln\left(\frac{200}{153}\right) \frac{n}{(\ln n)^3}\right) \longrightarrow 1. \quad (0.1)$$

The delicate point is the rounding phase of the independence-number threshold. We compare ordinary colorings with signed cocolorings whose classes have four consecutive sizes and may be declared either independent or complete. The sign choice creates an exquisitely phase-sensitive but uniform first-moment advantage, lowering the relevant root by order $n/(\ln n)^3$. To show that this lower root is attained, we derive an exact overlap identity and express its sign contribution through even subgraphs of a bipartite configuration model. The resulting second-moment estimate first produces a possibly rare cocoloring; a vertex-exposure argument then promotes that seed to a high-probability event without changing the leading root separation.

INTRODUCTION

Let $G_n \sim G(n, 1/2)$ be the random graph on $[n]$, with each edge present independently with probability $1/2$. The chromatic number $\chi(G)$ is the least number of independent sets in a partition of $V(G)$. The cochromatic number $\zeta(G)$ is the least number of parts in a partition in which every part induces either an empty graph or a complete graph. We say that an event holds with high probability if its probability tends to one as $n \rightarrow \infty$.

Erdős and Gimbel asked whether $\chi(G_n) - \zeta(G_n) \rightarrow \infty$ with high probability (Erdős and Gimbel, 1993, p. 263). The question was later restated by Gimbel (2016, Section 7.4, p. 100) and is catalogued as Erdős Problem 625 (Bloom, 2026). We prove the following quantitative form.

Theorem 1. *For $G_n \sim G(n, 1/2)$,*

$$\mathbb{P}\left(\chi(G_n) - \zeta(G_n) \geq \frac{(\ln 2)^2}{32} \ln\left(\frac{200}{153}\right) \frac{n}{(\ln n)^3}\right) \longrightarrow 1.$$

In particular, $\chi(G_n) - \zeta(G_n)$ diverges with high probability along the full sequence of integers n .

The full-sequence quantifier is essential. Near an independence-number jump, a rounding error of one vertex changes the available profile, so an estimate proved only on a dense set of phases does not automatically extend to every integer n . The argument below is therefore uniform in the complete phase parameter $\delta \in [0, 1)$, including sequences approaching either side of each jump.

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BACKGROUND AND RELATION TO EARLIER WORK.

The chromatic number of dense random graphs has been studied since the 1970s. Grimmett and McDiarmid (1975, Corollary 3) established the correct first-order lower scale and formulated the corresponding upper-bound problem. Matula (1987) developed an expose-and-merge approach, and Bollobás (1988, Theorem 4) proved the first-order asymptotic. Subsequent refinements include McDiarmid (1990), Panagiotou and Steger (2009, Theorem 1.2), and the sharp coloring-rate bounds of Heckel (2018, Theorem 1); see also Janson et al. (2000, Theorem 7.14 and Remark 7.15).

For the chromatic–cochromatic gap, Heckel (2024, Theorem 1) and, independently, Steiner (2024, Theorem 1.7) gave the first quantitative evidence toward divergence. Heckel proved that if a deterministic integer sequence $g(n)$ satisfies $\mathbb{P}(\chi(G_n) - \zeta(G_n) \leq g(n)) > 0.999$, then along an unbounded sequence n^* ,

$$g(n^*) > c \frac{\sqrt{n^*} \ln \ln n^*}{(\ln n^*)^3}.$$

This obstructs uniformly small high-probability upper bounds, but is not an all- n high-probability lower bound. Heckel also conjectured the scale $\Theta(n/(\ln n)^3)$ (Heckel, 2024, Conjecture 4).

The subsequent theorem of Heckel (2025, Theorem 1) proved that, for every fixed $\varepsilon > 0$,

$$n^{0.05+\varepsilon} \leq \mu_\alpha \leq n^{1-\varepsilon} \implies \chi(G_n) - \zeta(G_n) \geq n^{1-\varepsilon} \quad \text{with high probability.}$$

Thus the divergence question was settled on a phase-dependent set comprising roughly 95% of the integers; see Heckel (2025, Section 2).

Two ingredients of that work are strategic antecedents here. First, the one-partition factor 2^k and the need to count sign compatibility through the overlap matrix appear in Heckel (2025, Proposition 6). Section 5 rederives the first identity, while Section 6 replaces the pairwise upper bound by an exact component formula. Second, the rare-seed and concentration-amplification strategy follows Heckel (2025, Proposition 5 and the proof of Theorem 1). Sections 6–9 prove the signed second-moment estimate needed here, and Section 10 proves the quantitative amplifier. The four-size profile, its phase-uniform entropy certificate, the treatment of all remaining overlap regimes, the all- n conclusion, and the explicit constant in Theorem 1 are the contributions of this paper.

PROOF ROADMAP.

Sections 2–4 locate $\chi(G_n)$ from below without imposing a hidden profile restriction. Section 2 resolves the complete independence-number phase, Section 3 develops the continuous root geometry, and Section 4 turns that geometry into an unrestricted chromatic lower bound.

Section 5 constructs the four-size signed profile and proves the decisive first-moment separation of order $n/(\ln n)^3$. The rest of the paper shows that this formal root is genuinely populated. Section 6 derives the exact overlap and sign decomposition; Section 7 controls common whole classes; Section 8 sums every canonical high-cell skeleton; and Section 9 bounds the remaining conditional local and cycle attachments. Section 10 then amplifies the possibly rare second-moment seed to a high-probability cocoloring event. Section 11 subtracts the two locations. The roadmap records dependencies only; each asymptotic estimate is proved where it is used.

HOW TO READ THE PROOF.

The proof has three layers, which are best kept separate. The first layer is the *location calculation*: Sections 2–5 compare the ordinary coloring root with the signed four-size cocoloring root. This is where the constant $(\ln 2)^2 \ln(200/153)/32$ enters. The second layer is the *second-moment calculation*: Sections 6–9 prove that the signed first-moment advantage is not destroyed by overlaps. These sections contain most of the bookkeeping; their role is to bound one normalized second moment, not to move either root. The third layer is the *probability amplification*: Section 10 turns a possibly rare cocoloring

witness into a high-probability upper bound for $\zeta(G_n)$, and Section 11 compares it with the chromatic lower bound.

The main technical difficulty is that the second layer has to count overlaps without losing the phase-uniform constant from the first layer. To make the bookkeeping auditable, the overlap matrix is reduced in stages. Section 6 gives an exact sign identity. Section 7 removes common whole classes. Section 8 handles the high-multiplicity cells of the remaining overlap matrix. Section 9 handles the residual local factors and cycle-space attachments. Thus Lemmas 8.3 and 9.1 are not independent black boxes: Lemma 8.3 bounds the high skeletons before the residual graph is exposed, while Lemma 9.1 bounds what remains after one such skeleton has been fixed.

1. NOTATION AND ELEMENTARY FACTS

All logarithms are natural unless a base is displayed.

For an integer s , let

$$\mu_s(v) = \binom{v}{s} 2^{-\binom{s}{2}}, \quad \mu_s = \mu_s(n). \quad (1.1)$$

We use the following standard inequalities in their stated forms.

1. For integers $m \geq 1$,

$$\ln(m!) = m \ln m - m + \frac{1}{2} \ln(2\pi m) + O(1/m). \quad (1.2)$$

In particular, uniformly for $m \geq 0$, with $0 \ln 0 = 0$, the last two terms may be replaced by $O(\ln(m+1))$.

2. If a random variable Y is a function of r independent blocks and changing one block changes Y by at most one, then

$$\mathbb{P}(|Y - \mathbb{E}[Y]| \geq t) \leq 2 \exp(-2t^2/r). \quad (1.3)$$

We will use the corresponding one-sided bounds as well.

3. If $Z \geq 0$ and $0 < \mathbb{E}[Z] < \infty$, then

$$\mathbb{P}(Z > 0) \geq \frac{\mathbb{E}[Z]^2}{\mathbb{E}[Z^2]}. \quad (1.4)$$

4. If $X \sim \text{Bin}(m, 1/2)$, then

$$\mathbb{P}(X \leq m/4) \leq e^{-m/16}. \quad (1.5)$$

Indeed, exponential Markov with $t = \ln 3$ bounds the probability by $[3^{1/4}(2/3)]^m \leq e^{-m/16}$.

5. For every nonnegative random variable X and $a > 0$, $\mathbb{P}(X \geq a) \leq \mathbb{E}[X]/a$.

The first is Stirling's estimate, the second is McDiarmid's bounded-differences inequality, and the third is the zero-threshold case of Paley–Zygmund. The fourth is the only binomial-tail estimate used below; the fifth is Markov's inequality. The bounded-differences formulation is the one recorded by McDiarmid (1989, Theorem 3.1).

2. THE COMPLETE INDEPENDENCE-NUMBER PHASE

Define

$$\alpha_0 = 2 \log_2 n - 2 \log_2 \log_2 n + 2 \log_2(e/2) + 1, \quad \alpha = \lfloor \alpha_0 \rfloor, \quad \delta = \alpha_0 - \alpha. \quad (2.1)$$

Thus $0 \leq \delta < 1$. The estimates below are uniform on this entire interval, including sequences approaching either endpoint.

Lemma 2.1 (phase expansion).

There is a bounded continuous function $K: [0, 1] \rightarrow \mathbb{R}$ such that

$$\ln \mu_\alpha - K(\delta) = \delta \ln n + \left(\frac{2}{\ln 2} - \frac{1}{2} - \delta \right) \ln \ln n + O((\ln \ln n)^2 / \ln n). \quad (2.2)$$

In particular,

$$\mu_{\alpha+2} = n^{\delta-2+o(1)} \leq n^{-1+o(1)} = o(1), \quad (2.3)$$

and, with $a = \alpha - 2$,

$$\mu_a \geq cn^2(\ln n)^{2/\ln 2 - 5/2} \quad (2.4)$$

for an absolute $c > 0$ and all sufficiently large n .

Proof. Write

$$C = 1 + \ln(\ln 2) - \ln 2, \quad S = \ln n - \ln \ln n + C, \quad b = 1 - \delta.$$

Then $\alpha = 2S/\ln 2 + b$. Since $\alpha = O(\ln n)$, (1.2) gives

$$\ln(n)_\alpha = \alpha \ln n + \sum_{j=0}^{\alpha-1} \ln(1 - j/n) = \alpha \ln n + O(\alpha^2/n),$$

and hence

$$\ln \mu_\alpha = \ln(n)_\alpha - \ln(\alpha!) - \ln 2 \binom{\alpha}{2}.$$

Applying Stirling's formula to $\alpha!$ in this identity gives

$$\ln \mu_\alpha = \alpha \left(\ln n - \ln \alpha + 1 - \frac{\ln 2}{2}(\alpha - 1) \right) - \frac{1}{2} \ln(2\pi\alpha) + O(1/\ln n). \quad (2.5)$$

Expanding $\ln \alpha$ at $2\ln n/\ln 2$, uniformly in $b \in (0, 1]$, gives

$$\ln n - \ln \alpha + 1 - \frac{\ln 2}{2}(\alpha - 1) = \frac{\ln 2\delta}{2} + \frac{\ln \ln n - C - b\ln 2/2}{\ln n} + O((\ln \ln n)^2 / (\ln n)^2). \quad (2.6)$$

Put $q = \ln 2$ and $\ell = \ln \ln n$. Since $\alpha = 2(\ln n - \ell + C)/q + b$, multiplying the right side of (2.6) by α gives

$$\begin{aligned} \alpha \left\{ \frac{q\delta}{2} + \frac{\ell - C - bq/2}{\ln n} + O\left(\frac{\ell^2}{(\ln n)^2}\right) \right\} &= \delta \ln n + \left(\frac{2}{q} - \delta \right) \ell \\ &\quad + \delta C + \frac{q}{2}\delta(1 - \delta) - \frac{2C}{q} - (1 - \delta) + O\left(\frac{\ell^2}{\ln n}\right). \end{aligned}$$

Moreover,

$$-\frac{1}{2} \ln(2\pi\alpha) = -\frac{1}{2}\ell - \frac{1}{2} \ln(2\pi) - \frac{q}{2} + \frac{1}{2} \ln q + O\left(\frac{\ell}{\ln n}\right).$$

Substitution in (2.5) proves (2.2), with

$$K(\delta) = \delta C + \frac{\ln 2}{2} \delta(1 - \delta) - \frac{2C}{\ln 2} - (1 - \delta) - \frac{1}{2} \ln(2\pi) - \frac{\ln 2}{2} + \frac{1}{2} \ln(\ln 2). \quad (2.7)$$

The exact adjacent-size ratios are

$$\frac{\mu_{s+1}}{\mu_s} = \frac{n-s}{s+1} 2^{-s}, \quad \frac{\mu_{s-1}}{\mu_s} = \frac{s}{n-s+1} 2^{s-1}. \quad (2.8)$$

At $s = \alpha + O(1)$, the first is $\Theta(\ln n/n)$ and the second is $\Theta(n/\ln n)$, uniformly in δ . Thus

$$\mu_{\alpha+2} = \mu_\alpha \Theta\left(\frac{(\ln n)^2}{n^2}\right), \quad \mu_{\alpha-2} = \mu_\alpha \Theta\left(\frac{n^2}{(\ln n)^2}\right). \quad (2.8a)$$

Taking logarithms in the first relation of (2.8a) and using (2.2) gives

$$\ln \mu_{\alpha+2} = (\delta - 2) \ln n + O(\ln \ln n),$$

which proves (2.3), uniformly in the phase. For the lower adjacent size, $\delta(\ln n - \ln \ln n) \geq 0$, while K is bounded, so (2.2) gives

$$\ln \mu_\alpha \geq \left(\frac{2}{\ln 2} - \frac{1}{2}\right) \ln \ln n + \inf_{\delta \in [0,1]} K(\delta) + o(1).$$

Hence $\mu_\alpha \geq c_1 (\ln n)^{2/\ln 2 - 1/2}$. Taking logarithms in the second relation of (2.8a), or equivalently multiplying this lower bound by $\Theta(n^2/(\ln n)^2)$, proves (2.4). $\square$

Let $X_{\alpha+2}$ count independent sets of size $\alpha + 2$. Then $\mathbb{E}[X_{\alpha+2}] = \mu_{\alpha+2}$, and $\alpha(G_n) > \alpha + 1$ implies $X_{\alpha+2} \geq 1$. Thus (2.3) and Markov's inequality give

$$\mathbb{P}(\alpha(G_n) > \alpha + 1) = o(1). \quad (2.9)$$

3. CONTINUOUS PROFILE ROOTS

For a class size u , put

$$d_u = 2^{\binom{u}{2}} u!.$$

Write $u = \alpha - i$, so that i is the deficit from α . Let

$$S_+ = \{-1, 0, 1, 2, \dots\}, \quad S_4 = \{2, 3, 4, 5\}. \quad (3.1)$$

At finite n , S_+ is cut off at $i = \alpha - 1$, since class sizes are positive. For a real class count k , define

$$L_{\mathbf{k}} = n \ln n - n + k - \sum_i k_i \ln(k_i d_{\alpha-i}), \quad (3.2)$$

and let $L_S(n, k)$ be its maximum over real $k_i \geq 0$ satisfying

$$\sum_{i \in S} k_i = k, \quad \sum_{i \in S} (\alpha - i) k_i = n. \quad (3.3)$$

Set

$$T = \alpha - \frac{n}{k}. \quad (3.4)$$

The profile optimization has two jobs. It must locate the zero of the first-moment exponent, and it must compare two different supports at the same average class size. The affine part of the deficit weight

cancels under the fixed-mean constraint; only the curved part distinguishes the supports. The following lemma packages both facts, together with the uniform derivative needed to convert an entropy advantage into a root displacement.

Lemma 3.1 (root phase, derivative, and support comparison).

Let

$$s_0 = \alpha_0 - 1 - \frac{2}{\ln 2}, \quad T_0 = \alpha - s_0 = 1 + \frac{2}{\ln 2} - \delta. \quad (3.5)$$

For $S \in \{S_+, S_4\}$, and uniformly for $0 \leq c \leq \ln 2$, the equation $L_S(n, k) + ck = 0$ has a unique zero $r_{S,c}$ in the root corridor described below, and

$$\frac{n}{r_{S,c}} = s_0 + O(\ln \ln n / \ln n), \quad T = \alpha - \frac{n}{r_{S,c}} = T_0 + O(\ln \ln n / \ln n). \quad (3.6)$$

For every fixed $A > 0$, uniformly when $|n/k - s_0| \leq A \ln \ln n / \ln n$,

$$\frac{\partial}{\partial k} \{L_S(n, k) + ck\} = \frac{2}{\ln 2} (\ln n)^2 + O(\ln n \ln \ln n). \quad (3.7)$$

For two supports R, S , at the same feasible k ,

$$\frac{L_R(n, k) - L_S(n, k)}{k} = \mathcal{F}_{n,R}(T) - \mathcal{F}_{n,S}(T), \quad (3.8)$$

where

$$\mathcal{F}_{n,S}(T) = \max_{\substack{\sum p_i = 1 \\ \sum i p_i = T}} \left[- \sum_i p_i \ln p_i + \sum_i p_i h_n(i) \right]. \quad (3.8a)$$

To make the uniformity precise, fix

$$K_* = [2/\ln 2 - 1/10, 1 + 2/\ln 2 + 1/10] \subset (2, 5). \quad (3.9a)$$

Then, uniformly for $T \in K_*$,

$$\mathcal{F}_{n,S}(T) \longrightarrow \mathcal{F}_S(T) := \max_{\substack{\sum p_i = 1 \\ \sum i p_i = T}} \left[- \sum p_i \ln p_i - \frac{\ln 2}{2} \sum i^2 p_i \right]. \quad (3.9)$$

The effective Lagrange tilts are uniformly bounded on K_* . For S_4 , if $p_{n,i}(T)$ and $p_i(T)$ denote the finite and limiting optimizers, then

$$\sup_{T \in K_*} \max_{2 \leq i \leq 5} |p_{n,i}(T) - p_i(T)| = o(1), \quad \inf_{T \in K_*} \min_{2 \leq i \leq 5} p_{n,i}(T) > 0. \quad (3.9b)$$

All convergence and error terms are uniform in δ .

Proof. Put $s = n/k$ and $p_i = k_i/k$. Dividing (3.2) by k gives

$$\frac{L_S(n, n/s)}{n/s} = (s-1) \ln n - s + 1 + \ln s + \max_p \left[- \sum p_i \ln p_i - \sum p_i \ln d_{\alpha-i} \right]. \quad (3.10)$$

There is an exact affine-plus-curved decomposition

$$- \ln d_{\alpha-i} = A_n + B_n i + h_n(i), \quad (3.11)$$

where

$$A_n = -\ln d_\alpha, \quad B_n = \ln 2\alpha - \ln 2/2 + \ln \alpha, \quad (3.12)$$

and, for $i \geq 0$,

$$h_n(i) = -\frac{\ln 2}{2}i^2 + \sum_{r=0}^{i-1} \ln(1 - r/\alpha). \quad (3.13)$$

For $i = -1$, direct subtraction gives $h_n(-1) = -\ln 2/2 + \ln(\alpha/(\alpha + 1))$. Thus, for fixed i ,

$$h_n(i) = -\ln 2i^2/2 + O(i^2/\alpha), \quad (3.14)$$

while (3.13) also supplies the uniform bound $h_n(i) \leq -\ln 2i^2/2$. The target mean T_0 stays in the compact interval

$$\frac{2}{\ln 2} \leq T_0 \leq 1 + \frac{2}{\ln 2}. \quad (3.15)$$

To control the optimizing tilts uniformly, define

$$Z_{n,S}(\lambda) = \sum_{i \in S} \exp\{\lambda i + h_n(i)\}, \quad M_{n,S}(\lambda) = \frac{d}{d\lambda} \ln Z_{n,S}(\lambda),$$

with the finite- n cutoff understood for S_+ . Let Z_S and M_S denote the corresponding limits obtained by replacing $h_n(i)$ with $-(\ln 2)i^2/2$. Since K_* lies strictly inside the convex hull of each support, there is a fixed $\Lambda > 0$ such that

$$M_S(-\Lambda) < \min K_* < \max K_* < M_S(\Lambda) \quad (S = S_+, S_4).$$

On $[-\Lambda, \Lambda]$, the weights and their first two moments are dominated by a constant multiple of $\exp(C|i| - (\ln 2)i^2/2)$. Hence $Z_{n,S}$, $M_{n,S}$, and $M'_{n,S}$ converge uniformly to their limiting counterparts. In particular, the preceding strict inequalities also hold with $M_{n,S}$ for all sufficiently large n .

The derivative $M'_{n,S}$ is the variance of i under the tilted law. The limiting variance is positive and continuous on the compact interval $[-\Lambda, \Lambda]$, so uniform convergence gives a constant $v_* > 0$ such that

$$M'_{n,S}(\lambda) \geq v_* \quad (|\lambda| \leq \Lambda, S = S_+, S_4)$$

for all sufficiently large n . Thus the inverse mean maps are uniformly Lipschitz on K_* , and their finite- n tilts converge uniformly to the limiting tilts. This proves (3.9), and on the finite support S_4 it also proves (3.9b). The affine terms in (3.11) cancel under the fixed mean constraint, giving the exact difference identity (3.8).

Substitute (3.11) into (3.10). If $T = \alpha - s$, the scalar part at $s = s_0$ is, by Stirling,

$$s_0 \left(\ln n - \ln \alpha - \frac{\ln 2}{2} s_0 + \frac{\ln 2}{2} \right) - \ln n + T_0 + 1 + \ln s_0 + \frac{\ln 2}{2} T_0^2 - \frac{1}{2} \ln(2\pi\alpha) + O(1/\ln n). \quad (3.15a)$$

Write $q = \ln 2$ and $\ell = \ln \ln n$. Then

$$\frac{qs_0}{2} = \ln n - \ell + \ln q - q$$

exactly, while

$$\ln \alpha = \ln s_0 + \frac{T_0}{s_0} + O((\ln n)^{-2}), \quad \ln s_0 = \ell + q - \ln q + O(\ell/\ln n).$$

Consequently,

$$\begin{aligned}
s_0 \left(\ln n - \ln \alpha - \frac{q}{2} s_0 + \frac{q}{2} \right) &= s_0 \left(\ell - \ln q + \frac{3q}{2} - \ln \alpha \right) \\
&= s_0 \left(\ln s_0 - \ln \alpha + \frac{q}{2} + O(\ell/\ln n) \right) \\
&= -T_0 + \frac{q}{2} s_0 + O(\ell) \\
&= -T_0 + \ln n - \ell + \ln q - q + O(\ell).
\end{aligned}$$

Substituting this identity into (3.15a) cancels the terms $\ln n$ and T_0 ; the remaining terms are

$$-\ell + \ln q - q + 1 + \ln s_0 + \frac{q}{2} T_0^2 - \frac{1}{2} \ln(2\pi\alpha) + O(\ell) = O(\ell),$$

uniformly in the phase. The entropy term is $O(1)$ by the Gaussian tail just proved. Hence

$$L_S(n, n/s_0)/(n/s_0) = O(\ln \ln n). \quad (3.16)$$

Write $\Psi_S(s) = L_S(n, n/s)/(n/s)$, and let $\lambda_{n,S}(T)$ be the effective tilt of the maximizing profile at mean $T = \alpha - s$. Differentiating (3.10) by the envelope theorem gives

$$\Psi'_S(s) = \ln n - 1 + s^{-1} - B_n + \lambda_{n,S}(T).$$

The tilt is uniformly bounded on the fixed target interval, and

$$B_n = 2\ln n - \ln \ln n + O(1), \quad (3.17)$$

therefore (3.17) gives

$$\Psi'_S(s) = -\ln n + O(\ln \ln n) \quad (3.18)$$

uniformly in a fixed neighborhood of s_0 . For $\Psi_{S,c}(s) = \Psi_S(s) + c$, choose $C > 0$ so that

$$|\Psi_{S,c}(s_0)| \leq C \ln \ln n$$

for both supports and all $0 \leq c \leq \ln 2$. By (3.18), throughout a fixed corridor and for sufficiently large n , $\Psi'_{S,c} = \Psi'_S \leq -\frac{1}{2} \ln n$. If $A > 2C$, integration gives

$$\begin{aligned}
\Psi_{S,c} \left(s_0 - A \frac{\ln \ln n}{\ln n} \right) &\geq \left(\frac{A}{2} - C \right) \ln \ln n > 0, \\
\Psi_{S,c} \left(s_0 + A \frac{\ln \ln n}{\ln n} \right) &\leq \left(C - \frac{A}{2} \right) \ln \ln n < 0.
\end{aligned}$$

Strict monotonicity therefore gives the unique corridor zero and (3.6), uniformly for $0 \leq c \leq \ln 2$.

Throughout any fixed corridor $|s - s_0| \leq A \ln \ln n / \ln n$, the same equations give $\Psi_{S,c}(s) = O_A(\ln \ln n)$. Since $L_S + ck = k\Psi_{S,c}(s)$ and $ds/dk = -s/k$,

$$\frac{d}{dk} \{L_S + ck\} = \Psi_{S,c}(s) - s\Psi'_S(s) = \frac{2}{\ln 2} (\ln n)^2 + O_A(\ln n \ln \ln n). \quad (3.19)$$

This is (3.7). $\square$

4. A VALID UNRESTRICTED LOWER LOCATION FOR χ

Let $r_+(n) = r_{S_+,0}$ be the zero in Lemma 3.1, and put

$$k_\chi^- := \lfloor r_+(n) \rfloor - \lfloor \ln n \rfloor. \quad (4.1)$$

For an integer profile $\mathbf{k} = (k_i)_{-1 \leq i \leq \alpha-1}$, let $X_{\mathbf{k}}$ be the number of unordered proper colorings having exactly k_i classes of size $\alpha - i$. Direct enumeration gives

$$\mathbb{E}[X_{\mathbf{k}}] = \frac{n!}{\prod_i ((\alpha - i)!)^{k_i} k_i!} 2^{-\sum_i k_i \binom{\alpha-i}{2}}. \quad (4.2)$$

For an integer k , let $\mathcal{K}_{n,k,\alpha+1}$ be the set of such profiles satisfying $\sum_i k_i = k$ and $\sum_i (\alpha - i)k_i = n$, and define

$$E_{n,k,\alpha+1} := \sum_{\mathbf{k} \in \mathcal{K}_{n,k,\alpha+1}} \mathbb{E}[X_{\mathbf{k}}].$$

Thus $E_{n,k,\alpha+1}$ is the expected total number of unordered, exactly k -part, $(\alpha + 1)$ -bounded colorings. There are at most $(n + 1)^{\alpha+1} = \exp(O((\ln n)^2))$ bounded profiles. Applying (1.2) to (4.2), uniformly even when some $k_i = 0$, yields

$$\ln E_{n,k,\alpha+1} \leq L_+(n, k) + O((\ln n)^2). \quad (4.3)$$

By Lemma 3.1,

$$L_+(n, k_\chi^-) \leq -c(\ln n)^3, \quad (4.4)$$

so the expectation in (4.3) is $o(1)$. On the event (2.9), any coloring with at most k_χ^- nonempty parts can be refined, by splitting parts, to exactly k_χ^- nonempty independent parts, all of size at most $\alpha + 1$. Therefore

$$\mathbb{P}(\chi(G_n) \leq k_\chi^-) = o(1). \quad (4.5)$$

This is an unrestricted chromatic lower bound; the size cap has been removed probabilistically by (2.9).

5. THE FOUR-SIZE SIGNED FIRST-MOMENT ADVANTAGE

A *signed cocoloring* is a partition in which every class is declared either “independent” or “complete”; it is realized when each class induces the graph specified by its declaration. Thus the sign is a two-way declaration attached to a class, and the signed counts below count witnesses for ordinary cocolorings rather than a new graph invariant.

There are two logically separate tasks. First, restricting the deficits to $S_4 = \{2, 3, 4, 5\}$ must cost strictly less than $\ln 2$ per part. Second, the 2^k choices of signs must convert that strict inequality into a macroscopic separation of the two roots. Lemma 5.1 proves the first point; the derivative estimate from Lemma 3.1 then proves the second.

For $S \in \{S_+, S_4\}$, introduce the tilted weights, partition function, and mean map

$$w_i(\lambda) := \exp\left(\lambda i - \frac{\ln 2}{2} i^2\right), \quad Z_S(\lambda) := \sum_{i \in S} w_i(\lambda),$$

$$M_S(\lambda) := \frac{\sum_{i \in S} i w_i(\lambda)}{Z_S(\lambda)} = \frac{d}{d\lambda} \ln Z_S(\lambda).$$

The Gaussian factor makes both sums finite. Moreover, $M'_S(\lambda) = \text{Var}_{S,\lambda}(i) > 0$, so for every target mean T in the interior of the convex hull of S there is a unique tilt $\lambda_S(T)$ with $M_S(\lambda_S(T)) = T$. The optimizer in (3.9) is therefore

$$p_i = \frac{w_i(\lambda_S(T))}{Z_S(\lambda_S(T))} = \frac{e^{\lambda_S(T)i - \ln 2i^2/2}}{\sum_{j \in S} e^{\lambda_S(T)j - \ln 2j^2/2}}. \quad (5.1)$$

The same calculation gives the dual representation

$$\mathcal{F}_S(T) = \inf_{\lambda \in \mathbb{R}} \{\ln Z_S(\lambda) - \lambda T\} = \ln Z_S(\lambda_S(T)) - \lambda_S(T)T.$$

Define

$$D_4(\delta) = \mathcal{F}_{S_+}(T_0) - \mathcal{F}_{S_4}(T_0), \quad \gamma_4 = \ln \frac{200}{153}. \quad (5.2)$$

Lemma 5.1 (uniform entropy certificate).

For every $0 \leq \delta \leq 1$,

$$0 \leq D_4(\delta) < \ln \frac{153}{100}, \quad \ln 2 - D_4(\delta) > \gamma_4. \quad (5.3)$$

Proof. Fix $\delta \in [0, 1]$, put $T_0 = 1 + 2/\ln 2 - \delta$, and abbreviate

$$\lambda_4 := \lambda_{S_4}(T_0).$$

We first bracket this tilt. At $\lambda = 2\ln 2$, we make the reindexing completely explicit. The old summation index ranges over $i \in S_4 = \{2, 3, 4, 5\}$. Define the new index by

$$j := i - 2.$$

Then $i \mapsto j = i - 2$ is a bijection from $\{2, 3, 4, 5\}$ onto $\{0, 1, 2, 3\}$, with inverse $i = j + 2$. Thus the lower and upper bounds change from $2 \leq i \leq 5$ to $0 \leq j \leq 3$, respectively. Substituting $i = j + 2$ in the weight gives

$$\begin{aligned} w_{j+2}(2\ln 2) &= \exp\left(2\ln 2(j+2) - \frac{\ln 2}{2}(j+2)^2\right) \\ &= \exp\left(2\ln 2 - \frac{\ln 2}{2}j^2\right) = 4 \cdot 2^{-j^2/2}. \end{aligned}$$

Consequently, reindexing both sums in the definition of the mean—and only reindexing them, without discarding any term—gives

$$\begin{aligned} Z_{S_4}(2\ln 2) &= \sum_{i=2}^5 w_i(2\ln 2) = 4 \sum_{j=0}^3 2^{-j^2/2}, \\ \sum_{i=2}^5 i w_i(2\ln 2) &= 4 \sum_{j=0}^3 (j+2) 2^{-j^2/2} \\ &= 8 \sum_{j=0}^3 2^{-j^2/2} + 4 \sum_{j=0}^3 j 2^{-j^2/2}. \end{aligned}$$

Dividing the second equality by the first explains the additive constant 2 in the following formula:

$$\begin{aligned}
M_{S_4}(2\ln 2) &= 2 + \frac{\sum_{j=0}^3 j 2^{-j^2/2}}{\sum_{j=0}^3 2^{-j^2/2}} \\
&< 2 + \frac{\sum_{j \geq 0} j 2^{-j^2/2}}{\sum_{j \geq 0} 2^{-j^2/2}} < 2 + \frac{4}{5} < \frac{2}{\ln 2} \leq T_0.
\end{aligned} \tag{5.4}$$

For completeness, the first strict inequality in (5.4) has the following direct verification. Put

$$A = \sum_{j=0}^3 2^{-j^2/2}, \quad B = \sum_{j=0}^3 j 2^{-j^2/2}, \quad C = \sum_{j=4}^{\infty} 2^{-j^2/2}, \quad D = \sum_{j=4}^{\infty} j 2^{-j^2/2}.$$

Here $B/A \leq 3$, while $D/C \geq 4$. Since $A, C > 0$,

$$\frac{B+D}{A+C} - \frac{B}{A} = \frac{AD-BC}{A(A+C)} > 0.$$

Thus adjoining the terms whose new indices satisfy $j \geq 4$ strictly increases the weighted mean. For the next inequality, put $a_j = j 2^{-j^2/2}$. For $j \geq 4$,

$$\frac{a_{j+1}}{a_j} = \frac{j+1}{j} 2^{-(2j+1)/2} \leq \frac{1}{16},$$

and $a_4 = 1/64$. Hence

$$\sum_{j \geq 4} j 2^{-j^2/2} \leq \frac{1/64}{1 - 1/16} = \frac{1}{60}.$$

Writing $b = 2^{-1/2}$, retaining $j = 0, 1, 2$ in the denominator and $j = 0, 1, 2, 3$ in the numerator gives

$$\frac{\sum_{j \geq 0} j 2^{-j^2/2}}{\sum_{j \geq 0} 2^{-j^2/2}} < \frac{19b/16 + 1/2 + 1/60}{5/4 + b} < \frac{4}{5},$$

where the last inequality follows from $b < 71/100$. Finally, $2/3 < \ln 2 < 5/7$ gives $2 + 4/5 < 2/\ln 2$.

At $\lambda = 9\ln 2/2$, put $t = 2^{-1/8}$. The identity

$$\frac{9\ln 2}{2}i - \frac{\ln 2}{2}i^2 = \frac{\ln 2}{8}\{81 - (2i - 9)^2\}$$

gives

$$w_i(9\ln 2/2) = 2^{81/8} t^{(2i-9)^2}.$$

Thus the four weights, in the order $i = 2, 3, 4, 5$, are proportional to

$$t^{25}, \quad t^9, \quad t, \quad t.$$

The numerator of $M_{S_4}(9\ln 2/2) - 4$ is therefore

$$t - t^9 - 2t^{25} = t(1 - t^8 - 2t^{24}) = t/4 > 0, \tag{5.5}$$

and the denominator is positive. Hence this mean is greater than four, whereas $T_0 \leq 1 + 2/\ln 2 < 4$. Since M_{S_4} is strictly increasing, the two mean comparisons give

$$2\ln 2 < \lambda_4 < 9\ln 2/2. \tag{5.6}$$

For a tilt λ , define the omitted low- and high-deficit ratios

$$L(\lambda) := \frac{\sum_{i=-1}^1 w_i(\lambda)}{Z_{S_4}(\lambda)}, \quad H(\lambda) := \frac{\sum_{i \geq 6} w_i(\lambda)}{Z_{S_4}(\lambda)}.$$

Let $M_A(\lambda)$ denote the mean of i under the weights $w_i(\lambda)$ restricted to an index set A . Differentiation gives

$$\frac{d}{d\lambda} \ln L(\lambda) = M_{\{-1,0,1\}}(\lambda) - M_{S_4}(\lambda) < 0,$$

because the first mean is at most 1 and the second is at least 2. Similarly,

$$\frac{d}{d\lambda} \ln H(\lambda) = M_{\{6,7,\dots\}}(\lambda) - M_{S_4}(\lambda) > 0,$$

because the two means are at least 6 and at most 5, respectively. Direct summation and Gaussian-tail bounds give

$$L(2\ln 2) < \frac{51}{100}, \quad H(3\ln 2) < \frac{1}{50}, \quad L(3\ln 2) < \frac{3}{25}, \quad H(9\ln 2/2) < \frac{1}{4}. \quad (5.7)$$

Here are explicit checks. At $2\ln 2$, with $b = 2^{-1/2}$,

$$L(2\ln 2) = \frac{b + 1/4 + b/16}{1 + b + 1/4 + b/16} < 51/100.$$

At $\lambda = 3\ln 2$,

$$w_i(3\ln 2) = \exp\left((\ln 2) \left(3i - \frac{i^2}{2}\right)\right) = 2^{9/2} 2^{-(i-3)^2/2}.$$

Set $b = 2^{-1/2}$. After canceling the common factor $2^{9/2}$, the weight at index i is $b^{(i-3)^2}$. Thus the four S_4 -weights, for $i = 2, 3, 4, 5$, are

$$b, \quad 1, \quad b, \quad b^4,$$

and

$$2^{-9/2} Z_{S_4}(3\ln 2) = 1 + 2b + b^4 = \frac{5}{4} + 2b.$$

For the omitted low indices $i = -1, 0, 1$, the weights are

$$b^{16}, \quad b^9, \quad b^4 = \frac{1}{256}, \quad \frac{b}{16}, \quad \frac{1}{4}.$$

For the omitted high indices $i \geq 6$, they are

$$b^9 + b^{16} + \sum_{q \geq 5} b^{q^2}.$$

The first term in the remaining sum is $b^{25} = b/4096$, and the ratio between successive terms is at most $b^{11} < 1/32$. Therefore

$$\sum_{q \geq 5} b^{q^2} \leq \frac{b^{25}}{1 - 1/32} < \frac{1}{3968}.$$

It follows that

$$L(3\ln 2) = \frac{1/256 + b/16 + 1/4}{5/4 + 2b} < \frac{3}{25},$$

and

$$H(3\ln 2) \leq \frac{b/16 + 1/256 + 1/3968}{5/4 + 2b} < \frac{1}{50},$$

where the final inequalities follow from $7/10 < b < 71/100$.

Finally, at $9\ln 2/2$ the normalized exponent is $(2i - 9)^2$. Dividing all weights by the common S_4 weight t , the high indices $i = 6, 7, 8$ contribute t^8, t^{24}, t^{48} . From $i = 8$ onward the ratio between successive terms is at most $1/16$, so the tail beginning with t^{48} is less than $1/60$. The normalized S_4 denominator is $2 + t^8 + t^{24}$. Hence $H(9\ln 2/2) < 1/4$ follows from

$$3(t^8 + t^{24}) + 4/60 < 2, \quad (5.8)$$

because $t^8 = 1/2$ and $t^{24} = 1/8$.

If $\lambda_4 \leq 3\ln 2$, monotonicity and (5.7) give $L(\lambda_4) + H(\lambda_4) < 53/100$; if $\lambda_4 \geq 3\ln 2$, they give $L(\lambda_4) + H(\lambda_4) < 37/100$. By the dual representation above,

$$\mathcal{F}_{S_4}(T_0) = \ln Z_{S_4}(\lambda_4) - \lambda_4 T_0,$$

whereas evaluating the S_+ dual function at the same parameter gives

$$\mathcal{F}_{S_+}(T_0) \leq \ln Z_{S_+}(\lambda_4) - \lambda_4 T_0.$$

Subtracting these two relations yields

$$\begin{aligned} D_4(\delta) &\leq \ln \frac{Z_{S_+}(\lambda_4)}{Z_{S_4}(\lambda_4)} \\ &= \ln(1 + L(\lambda_4) + H(\lambda_4)) < \ln(153/100). \end{aligned}$$

Since S_4 is a subset of S_+ , the loss is nonnegative. Subtracting from $\ln 2$ completes the proof. $\square$

The following 2^k gain is the one-partition identity isolated by Heckel (2025, Proposition 6). We repeat its short argument because it is an input to the new root calculation. For a fixed partition into k classes, assigning each class the declaration “independent” or “complete” multiplies its first moment by 2^k : every declaration prescribes all internal edge bits, and all 2^k declarations have probability $2^{-\sum \binom{n}{2}}$. The declarations are disjoint here because all four allowed class sizes tend to infinity (in particular, there are no singleton classes). Let $r_4^{\text{co}} = r_{S_4, \ln 2}$ be the unique corridor zero of

$$L_{S_4}(n, k) + \ln 2k. \quad (5.9)$$

Put

$$\Phi_n(k) := L_{S_4}(n, k) + \ln 2k.$$

Then $\Phi_n(r_4^{\text{co}}) = 0$ by definition, while at $k = r_+$, Lemma 3.1 and (5.2) give

$$L_{S_4}(n, r_+) + \ln 2r_+ = r_+ \{\ln 2 - D_4(\delta) + o(1)\}. \quad (5.10)$$

Both roots lie in the corridor (3.6), so the whole interval between them is inside a fixed corridor. The mean-value theorem therefore gives a point $\xi_n \in (r_4^{\text{co}}, r_+)$ such that

$$\Phi_n(r_+) - \Phi_n(r_4^{\text{co}}) = \Phi'_n(\xi_n)(r_+ - r_4^{\text{co}}).$$

Now $r_+ = (\ln 2/2 + o(1))n/\ln n$, and (3.7), uniformly at ξ_n , gives

$$\Phi'_n(\xi_n) = \frac{2}{\ln 2}(\ln n)^2 + O((\ln n)(\ln \ln n)).$$

Substituting this and (5.10) into the preceding identity yields

$$r_+ - r_4^{\text{co}} = \left(\frac{(\ln 2)^2}{4} \{\ln 2 - D_4(\delta)\} + o(1) \right) \frac{n}{(\ln n)^3}. \quad (5.11)$$

In particular, for all sufficiently large n ,

$$r_+ - r_4^{\text{co}} \geq \frac{(\ln 2)^2 \gamma_4}{8} \frac{n}{(\ln n)^3}. \quad (5.12)$$

Choose the midpoint integer

$$k_{\text{co}} = \left\lceil \frac{r_4^{\text{co}} + r_+}{2} \right\rceil. \quad (5.13)$$

We call the resulting profile the *exact midpoint profile*: its total number of parts is the rounded midpoint between the signed four-size root r_4^{co} and the unrestricted chromatic root r_+ . The type counts are chosen next so that both finite- n conservation constraints hold exactly.

At this exact k , let $p_i^{(n)}$ be the exact finite- n maximizer of L_{S_4} . Let $\lambda_{n,\text{def}}$ denote the Lagrange multiplier for the deficit-mean constraint, with the sign convention that its contribution is $e^{\lambda_{n,\text{def}} i}$. The Lagrange equations and the decomposition (3.11) then give

$$p_i^{(n)} \propto d_{\alpha-i}^{-1} e^{\lambda_{n,\text{def}} i} \propto e^{\widehat{\lambda}_n i + h_n(i)}, \quad \widehat{\lambda}_n := B_n + \lambda_{n,\text{def}}. \quad (5.14)$$

Thus $\lambda_{n,\text{def}}$ is the multiplier before the affine term $B_n i$ is absorbed, while $\widehat{\lambda}_n$ is the effective tilt in the limiting Gaussian coordinates. (If a size-coordinate multiplier τ_n is used instead, then $\lambda_{n,\text{def}} = -\tau_n$.) The target deficit mean is $\alpha - n/k_{\text{co}}$. Lemma 3.1 makes these proportions converge uniformly to (5.1), with $\widehat{\lambda}_n \rightarrow \lambda_4(T_0)$. Equation (5.6) also shows directly that the four limiting weights differ by at most a fixed factor. Thus

$$\min_{2 \leq i \leq 5} p_i^{(n)} \geq c_p > 0 \quad (5.15)$$

uniformly in the phase.

Round $k_{\text{co}} p_i^{(n)}$ to preliminary integers $\widetilde{k}_i$. Define the signed constraint errors

$$e_0 = \sum_i \widetilde{k}_i - k_{\text{co}}, \quad e_1 = \sum_i i \widetilde{k}_i - (\alpha k_{\text{co}} - n),$$

so $e_0, e_1 = O(1)$, and add

$$\Delta k_2 = e_1 - 3e_0, \quad \Delta k_3 = 2e_0 - e_1. \quad (5.16)$$

Indeed, $\Delta k_2 + \Delta k_3 = -e_0$ and $2\Delta k_2 + 3\Delta k_3 = -e_1$. This enforces both constraints exactly, changes only $O(1)$ counts, and preserves positivity by (5.15). Put $\mathbf{k}^* = (k_{\text{co}} p_i^{(n)})_{i=2}^5$ and let $\Delta^{\text{tot}} = \mathbf{k} - \mathbf{k}^*$ be the total rounding and correction displacement. It has bounded norm and satisfies both homogeneous constraint equations, because $\mathbf{k}^*$ and $\mathbf{k}$ obey the same two constraints. Thus it is tangent to the feasible affine plane. At the exact optimizer the linear term vanishes, while the Hessian of the entropy term is $-\text{diag}(1/k_i)$. Hence

$$L_{\mathbf{k}^* + \Delta^{\text{tot}}} - L_{\mathbf{k}^*} = O\left(\sum_i \frac{(\Delta_i^{\text{tot}})^2}{k_i^*}\right) = O(1/k_{\text{co}}).$$

The preceding rounding and correction have already produced the exact integer vector

$$\mathbf{k} = (k_2, k_3, k_4, k_5), \quad u_i = \alpha - i, \quad k_i = \Theta(n/\ln n). \quad (5.17)$$

Applying (1.2) to the factorials in the exact signed first moment of this vector gives

$$\mathbb{E}[Z_{\mathbf{k}}^{\text{sgn}}] = 2^{k_{\text{co}}} \frac{n!}{\prod_i (u_i!)^{k_i} k_i!} 2^{-\sum_i k_i \binom{u_i}{2}}, \quad (5.18)$$

and

$$\ln \mathbb{E}[Z_{\mathbf{k}}^{\text{sgn}}] = L_{S_4}(n, k_{\text{co}}) + \ln 2k_{\text{co}} + O(\ln n) \geq c_Z k_{\text{co}} \quad (5.19)$$

for a phase-independent $c_Z > 0$. Indeed, the midpoint is at distance at least $(r_+ - r_4^{\text{co}})/2 - O(1)$ to the right of r_4^{co} . Integrating the positive derivative (3.7) over that interval and using (5.12) gives the scale calculation

$$\frac{n}{(\ln n)^3} \cdot (\ln n)^2 \asymp \frac{n}{\ln n} \asymp k_{\text{co}}.$$

Consequently,

$$L_{S_4}(n, k_{\text{co}}) + \ln 2k_{\text{co}} \geq c_Z k_{\text{co}},$$

after decreasing c_Z if necessary. Every u_i tends to infinity, so the independent and complete declarations of one part are disjoint. Thus $Z_{\mathbf{k}}^{\text{sgn}} > 0$ implies an actual cocoloring with k_{co} parts.

Finally, (4.1), (5.12), and (5.13) give

$$k_{\chi}^- - k_{\text{co}} \geq \left(\frac{(\ln 2)^2 \gamma_4}{16} - o(1) \right) \frac{n}{(\ln n)^3}. \quad (5.20)$$

6. EXACT SIGNED SECOND-MOMENT REPRESENTATION

For two partitions sharing exactly ℓ whole classes, Heckel (2025, Proposition 6) bounded the number of joint sign declarations by $2^{2k-\ell}$, thereby transferring selected ordinary-coloring overlap bounds to cocolorings. Lemma 6.1 below does not import that inequality. It keeps the entire overlap matrix and counts the compatible signs exactly: compatibility is controlled by the components of the graph of cells of size at least two. Whole common classes appear as one special case of this component formula.

Temporarily label all slots within each of the four types. This multiplies the unordered witness by the deterministic factor $\prod_i k_i!$ and hence does not change its normalized second moment. Choose two independent uniform ordered partitions of the profile. If their slots are indexed by a and b , put

$$r_{ab} = |V_a \cap W_b|. \quad (6.1)$$

The overlap matrix has the exact law

$$p(r) = \frac{\prod_a s_a! \prod_b t_b!}{n! \prod_{a,b} r_{ab}!}, \quad (6.2)$$

where here both degree lists are the multiset given by (5.17). Equivalently, place s_a stubs at each row slot and t_b stubs at each column slot and take a uniform perfect matching of the two n -stub sets.

Let $H(r)$ be the simple bipartite graph whose edges are cells with $r_{ab} \geq 2$, omitting isolated slot vertices. Write

$$W = \sum_{a,b} \binom{r_{ab}}{2}, \quad \beta(H) = |E(H)| - |V(H)| + c(H). \quad (6.3)$$

Lemma 6.1 (exact sign sum).

The normalized local factor of the pair of partitions is

$$A_{\zeta}(r) = 2^{W+c(H)-|V(H)|} = \left(\prod_{a,b} g(r_{ab}) \right) 2^{\beta(H)}, \quad (6.4)$$

where

$$g(0) = g(1) = g(2) = 1, \quad g(x) = 2^{\binom{x}{2}-1} \quad (x \geq 3). \quad (6.5)$$

Consequently

$$\frac{\mathbb{E}[(Z_{\mathbf{k}}^{\text{sgn}})^2]}{\mathbb{E}[Z_{\mathbf{k}}^{\text{sgn}}]^2} = \sum_{r \in \mathcal{R}(\mathbf{s}, \mathbf{t})} p(r) A_{\zeta}(r). \quad (6.6)$$

where $\mathcal{R}(\mathbf{s}, \mathbf{t})$ is the finite set of nonnegative integer matrices with the prescribed row margins $\mathbf{s}$ and column margins $\mathbf{t}$.

Proof. Let $B = \sum_a \binom{s_a}{2}$, which is the number of internal edge bits prescribed by one partition. A row sign and a column sign are compatible on a cell of size at least two exactly when they agree. Therefore signs must be constant on every component of H . There are

$$2^{2k_{\text{co}} - |V(H)| + c(H)}$$

compatible pairs of sign assignments, including the free signs on slots isolated from H . For each compatible pair, W edge bits are prescribed twice, so the probability of all constraints is $2^{-(2B-W)}$. Division by the square of the one-partition signed probability $(2^{k_{\text{co}}} 2^{-B})^2$ gives

$$\frac{2^{2k_{\text{co}} - |V(H)| + c(H)} 2^{-(2B-W)}}{(2^{k_{\text{co}}} 2^{-B})^2} = 2^{W + c(H) - |V(H)|},$$

which is the first expression in (6.4).

Formula (6.4) separates two effects that will be estimated by different methods. The product of the $g(r_{ab})$ is purely local and is trivial for cell sizes at most two; the factor $2^{\beta(H)}$ is topological and counts the independent cycle choices of the support graph.

For every support edge, split $\binom{r}{2}$ as $(\binom{r}{2} - 1) + 1$. Since $|E| - |V| + c = \beta$,

$$\prod_{a,b} g(r_{ab}) = 2^{W - |E(H)|},$$

and multiplying by $2^{\beta(H)}$ gives the second expression. Averaging over (6.2) proves (6.6). $\square$

The identity

$$2^{\beta(H)} = \#\{F \subseteq E(H) : \deg_F(v) \text{ is even for every } v\} \quad (6.7)$$

will be used repeatedly. It is the elementary fact that the even subgraphs form the binary cycle space of dimension $\beta(H)$.

We also need one exact configuration-model estimate.

Lemma 6.2 (joint prescribed-cell bound).

In a bipartite configuration model of total degree m_0 , with row degrees d_a and column degrees d'_b , prescribe distinct cells D and demands $x_{ab} \geq 1$. Put

$$x = \sum_D x_{ab}, \quad D_a = \sum_b x_{ab}, \quad D'_b = \sum_a x_{ab}.$$

In the feasible case

$$x \leq m_0, \quad D_a \leq d_a \quad (\forall a), \quad D'_b \leq d'_b \quad (\forall b),$$

we have

$$\mathbb{P}(r_{ab} \geq x_{ab} \text{ for every } ab \in D) \leq \frac{\prod_a (d_a)_{D_a} \prod_b (d'_b)_{D'_b}}{(m_0)_x \prod_{ab \in D} x_{ab}!}. \quad (6.8)$$

If any feasibility condition fails, the prescribed event is empty and is handled separately. In particular, in all cases with $m_0 > 0$, with $e := \exp(1)$ and $\theta_{ab} = e d_a d'_b / m_0$, its probability is at most

$$\prod_{ab \in D} \frac{\theta_{ab}^{x_{ab}}}{x_{ab}!}. \quad (6.9)$$

Proof. If a feasibility condition fails, no matching realizes all demands. In the feasible case, for each row a , assigning disjoint row stubs to its demanded cells gives $(d_a)_{D_a} / \prod_b x_{ab}!$ choices. The analogous column choice gives $(d'_b)_{D'_b} / \prod_a x_{ab}!$ choices, and there are $x_{ab}!$ bijections between the selected stubs inside cell ab . Thus the total number of demand witnesses is

$$\frac{\prod_a (d_a)_{D_a} \prod_b (d'_b)_{D'_b}}{\prod_{ab \in D} x_{ab}!}.$$

Any fixed witness occurs with probability $1/(m_0)_x$, because after its x prescribed pairs are exposed the remaining matching is uniform on the unused stubs. A union bound proves (6.8).

It remains to simplify the global denominator. For $1 \leq x \leq m_0$,

$$\binom{m_0}{x} = \prod_{j=0}^{x-1} \frac{m_0 - j}{x - j} \geq \left(\frac{m_0}{x}\right)^x, \quad x! \geq \left(\frac{x}{e}\right)^x.$$

The factorial inequality follows, for example, from $\sum_{j=1}^x \ln j \geq \int_1^x \ln t \, dt = x \ln x - x + 1$. Consequently

$$(m_0)_x = x! \binom{m_0}{x} \geq (m_0/e)^x, \quad (6.10)$$

with the case $x = 0$ immediate. Combining (6.10) with $(d)_r \leq d^r$ proves (6.9). Notice that (6.8) retains the single global falling factorial before (6.9) is taken. $\square$

7. EXACT PARTIAL DIAGONALS

Purpose of this section. The exact overlap representation of Section 6 still permits two partitions to share whole classes. The definitions below assign a nonnegative marked weight to each such common subprofile. We prove that the total marked contribution is $1 + o(1)$, uniformly through the phase. This partial-diagonal estimate is the input used in Section 8 to control the dense transportation sum. The counting identities in (7.1)–(7.4) are finite- n identities; asymptotic estimates are explicitly labeled below.

A *partial diagonal* is a specified collection of whole classes that the two partitions have in common. It is marked: one overlap may contain several such collections and then contributes to every corresponding nonnegative marked term. This intentional overcount is what makes the subsequent summation an upper-bound device.

For a selected common subprofile $\ell = (\ell_i)$, put

$$\ell_\bullet = \sum_i \ell_i, \quad m = \sum_i u_i \ell_i, \quad b_\ell = \sum_i \binom{u_i}{2} \ell_i. \quad (7.1)$$

Here $0 \leq \ell_i \leq k_i$ coordinatewise. Since the full profile has $\sum_i u_i k_i = n$, this automatically gives $m \leq n$; thus every factorial in the counting formula below has its ordinary combinatorial meaning.

Exact marked decomposition. The signed first moment of such partial partitions occupying any m vertices of $[n]$ is

$$Y_\ell^{\text{sgn}} = 2_\bullet^\ell \frac{n!}{(n-m)! \prod_i (u_i!)^{\ell_i} \ell_i!} 2^{-b_\ell}. \quad (7.2)$$

For each type i , choose the ℓ_i marked slots in the first partition and the ℓ_i marked slots in the second, giving $\binom{k_i}{\ell_i}^2$ choices. The common blocks themselves have the signed first-moment weight Y_ℓ^{sgn} . In the normalized second moment they are counted once rather than independently in each marginal, so their contribution is the reciprocal of this weight. Therefore the exact *marked* common-subprofile weight is

$$D(\ell) = \frac{\prod_i \binom{k_i}{\ell_i}^2}{Y_\ell^{\text{sgn}}}. \quad (7.3)$$

Different marked choices need not be disjoint: an overlap containing several common blocks contributes to each corresponding marked choice. The sums below are sums of these nonnegative marked weights, so no disjointness is used. In particular, $D(0) = 1$ and $D(k) = 1/\mathbb{E}[Z_k^{\text{sgn}}]$. To add one common block of type i , first note that

$$\frac{\binom{k_i}{\ell_i+1}^2}{\binom{k_i}{\ell_i}^2} = \frac{(k_i - \ell_i)^2}{(\ell_i + 1)^2},$$

while (7.2) gives

$$\frac{Y_{\ell+e_i}^{\text{sgn}}}{Y_\ell^{\text{sgn}}} = \frac{2(n-m)_{u_i}}{u_i! (\ell_i + 1) 2^{\binom{u_i}{2}}} = \frac{2\mu_{u_i}(n-m)}{\ell_i + 1}.$$

Dividing these two ratios gives

$$\frac{D(\ell + e_i)}{D(\ell)} = \frac{(k_i - \ell_i)^2}{2(\ell_i + 1)\mu_{u_i}(n-m)}. \quad (7.4)$$

At the opposite endpoint, put $h = k - \ell$, $v = \sum_i u_i h_i$, and let $Z_h^{\text{sgn}}(v)$ be the signed first moment of a complete profile h on v vertices. Exact factorial cancellation gives

$$D(k - h) = \frac{B(h)}{\mathbb{E}[Z_k^{\text{sgn}}]}, \quad B(h) = \left(\prod_i \binom{k_i}{h_i} \right) Z_h^{\text{sgn}}(v), \quad (7.5)$$

and

$$\frac{\binom{k_i}{h_i+1}}{\binom{k_i}{h_i}} = \frac{k_i - h_i}{h_i + 1}, \quad \frac{Z_{h+e_i}^{\text{sgn}}(v + u_i)}{Z_h^{\text{sgn}}(v)} = \frac{2\mu_{u_i}(v + u_i)}{h_i + 1}.$$

Therefore

$$\frac{B(h + e_i)}{B(h)} = \frac{2(k_i - h_i)\mu_{u_i}(v + u_i)}{(h_i + 1)^2}. \quad (7.6)$$

Equations (7.2)–(7.6) are exact finite- n counting identities. We now use them only through uniform asymptotic bounds on the three ranges of selected mass.

In Heckel's admissible-phase argument, the analogous control of partial profiles enters through the tame-coloring estimates of Heckel and Panagiotou (2023, Lemmas 5.1–5.3, 6.3–6.5, and 7.20), as used in Heckel (2025, Proposition 5); their hypotheses are the source of the lower μ_α -window in that theorem.

The next lemma is the replacement needed here. It is proved directly for the four-size signed profile, including both corners and every intermediate mass, and no tame-profile theorem is invoked.

Lemma 7.1 (all common subprofiles).

For the exact midpoint profile,

$$\sum_{0 \leq \ell_i \leq k_i} D(\ell) = 1 + o(1). \quad (7.7)$$

The proof has three ranges, classified by the vertex mass occupied by the marked common classes. The recurrence (7.4) controls the empty corner, the continuous rate function controls the central range, and the reverse recurrence (7.6) controls the full corner. The ranges overlap only at their boundaries and together exhaust every common subprofile.

Proof: the empty corner. Let

$$\eta = \frac{\ln \ln n}{32 \ln n}, \quad \xi_i = \frac{k_i^2}{2\mu_{u_i}(n)}, \quad \Xi_{\text{empty}} = \sum_i \xi_i. \quad (7.8)$$

Equations (2.4), (2.8), and $k_i = \Theta(n/\ln n)$ imply

$$\Xi_{\text{empty}} = O((\ln n)^{-(2/\ln 2 - 1/2)}) = o(1). \quad (7.9)$$

If the final selected mass is at most ηn , every intermediate mass m' in (7.4) satisfies

$$\frac{\mu_{u_i}(n)}{\mu_{u_i}(n - m')} \leq (1 - 2\eta)^{-u_i} \leq (\ln n)^{3/8}. \quad (7.10)$$

Iterating (7.4), in any order, yields

$$D(\ell) \leq \prod_i \frac{((\ln n)^{3/8} \xi_i)^{\ell_i}}{\ell_i!}.$$

Thus

$$1 \leq \sum_{m \leq \eta n} D(\ell) \leq \exp((\ln n)^{3/8} \Xi_{\text{empty}}) = 1 + o(1). \quad (7.11)$$

Proof: the central range. Write

$$\begin{aligned} p_i &= k_i/k_{\text{co}}, & y_i &= \ell_i/k_{\text{co}}, & Y &= \sum_i y_i, & I &= \sum_i i y_i, \\ z_i &= p_i - y_i, & R &= \sum_i z_i = 1 - Y, & I_r &= \sum_i i z_i = T - I. \end{aligned} \quad (7.12)$$

The residual vertex fraction is

$$\rho = \frac{n - m}{n} = R + \frac{I - TY}{\alpha - T}. \quad (7.13)$$

Uniform asymptotic estimate: Stirling control. Before applying Stirling, (7.2) and (7.3) give the exact logarithmic identity

$$\begin{aligned} \ln D(\ell) = 2 \sum_i \{ \ln(k_i!) - \ln(\ell_i!) - \ln((k_i - \ell_i)!) \} \\ - \ell_\bullet \ln 2 - \ln(n!) + \ln((n - m)!) + \sum_i \ell_i \ln(u_i!) + \sum_i \ln(\ell_i!) + (\ln 2)b_\ell. \end{aligned}$$

Applying Stirling to this identity, with the convention $0 \ln 0 = 0$, and using $n = k_{\text{co}}(\alpha - T)$, gives uniformly

$$\begin{aligned} \ln D(\ell) = n\rho \ln \rho \\ + k_{\text{co}} \sum_i \{ 2p_i \ln p_i - 2(p_i - y_i) \ln(p_i - y_i) - y_i \ln y_i - y_i + y_i E_i \} + O(\ln n), \end{aligned} \quad (7.14)$$

where

$$E_i = \ln k_{\text{co}} + \ln(u_i!) + u_i - u_i \ln n + \ln 2 \binom{u_i}{2} - \ln 2. \quad (7.15)$$

Since $u_{i+1} = u_i - 1$, exact subtraction in (7.15) gives

$$\begin{aligned} E_{i+1} - E_i &= \ln n - \ln(\alpha - i) - (\ln 2)(\alpha - i) + \ln 2 - 1 \\ &= -\frac{\ln 2}{2} \alpha + O(1). \end{aligned} \quad (7.16)$$

For the second line, substitute $(\ln 2)\alpha/2 = \ln n - \ln \ln n + O(1)$ from (2.1) and $\ln(\alpha - i) = \ln \ln n + O(1)$, uniformly for $2 \leq i \leq 5$.

Let $\bar{E} = \sum p_i E_i$. Applying Stirling to the complete moment (5.18) gives

$$-\frac{1}{k_{\text{co}}} \ln \mathbb{E}[Z_k^{\text{sgn}}] = \sum_i p_i \ln p_i - 1 + \bar{E} + o(1). \quad (7.17)$$

The positive margin (5.19) bounds $\bar{E}$ above. Indeed, (7.17) and $\ln \mathbb{E}[Z_k^{\text{sgn}}] \geq c_Z k_{\text{co}}$ give $\bar{E} = O(1)$ from above, uniformly in the phase. Since the support consists of four consecutive indices, summing (7.16) over at most three steps and using $\sum_i p_i(i - T) = 0$ gives

$$E_i = \bar{E} - \frac{\ln 2 \alpha}{2} (i - T) + O(1) \quad (2 \leq i \leq 5).$$

Multiplying by y_i and summing yields

$$\sum_i y_i E_i \leq \frac{\ln 2 \alpha}{2} (TY - I) + O(Y). \quad (7.18)$$

Indeed,

$$\sum_i y_i E_i = Y\bar{E} - \frac{(\ln 2)\alpha}{2} (I - TY) + O(Y).$$

Only an upper bound is needed in (7.18), and $\bar{E} \leq C$ uniformly; therefore $Y\bar{E} \leq CY$, which explains its absorption into $O(Y)$. The remaining entropy expression in (7.14) is $O(Y \ln(e/Y))$, uniformly because every p_i is bounded below. If $\rho \geq 1/32$, then $R \geq 1/64$ for large n . Equation (7.13) and $2 \leq i, T \leq 5$ give

$$|\rho - R| = \frac{|I - TY|}{\alpha - T} = O(Y/\alpha).$$

The derivative of $x \ln x$ is bounded on the interval between ρ and R . The mean-value theorem, together with $n = k_{\text{co}}(\alpha - T)$ and $|R \ln R| = O(1 - R) = O(Y)$, therefore gives

$$n\rho \ln \rho = k_{\text{co}}\alpha R \ln R + O(k_{\text{co}}Y). \quad (7.19)$$

Consequently

$$\ln D(\ell) \leq k_{\text{co}}\alpha \Phi_T(z) + Ck_{\text{co}}Y \ln(e/Y) + C \ln n, \quad (7.20)$$

where

$$\Phi_T(z) = R \ln R + \frac{\ln 2}{2}(I_r - TR). \quad (7.21)$$

Uniform asymptotic estimate: rate negativity. This rate is uniformly negative away from its two corners. Indeed,

$$I_r - TR \leq (5 - T)R, \quad I_r - TR = \sum_i (T - i)y_i \leq (T - 2)(1 - R), \quad (7.22)$$

and hence

$$\Phi_T \leq R\{\ln R + \ln 2(5 - T)/2\}, \quad \Phi_T \leq R \ln R + \ln 2(T - 2)(1 - R)/2. \quad (7.23)$$

Here is the numerical check used in this split. On $1/64 \leq R \leq 47/100$, add $Y/5000 = (1 - R)/5000$ to the first bound in (7.23). The resulting function

$$R \ln R + \frac{\ln 2}{2}(5 - T)R + \frac{1 - R}{5000}$$

is convex in R . Its largest possible coefficient occurs at $T = 2/\ln 2$. Using

$$0.6931 < \ln 2 < 0.6932, \quad \ln 64 > 4.1588, \quad \ln(100/47) > 0.7550,$$

its values at the two endpoints are bounded above by

$$-0.0533 \quad \text{at } R = 1/64, \quad -0.0102 \quad \text{at } R = 47/100.$$

Convexity therefore gives $\Phi_T \leq -Y/5000$ throughout this interval.

On $47/100 \leq R \leq 1$, use the second bound in (7.23) and $T \leq 1 + 2/\ln 2$. After adding $Y/200$, it is at most

$$R \ln R + (1 - \ln 2/2 + 1/200)(1 - R)$$

This function is convex; the same numerical bounds give a value below -0.0058 at $R = 47/100$, while its value at $R = 1$ is 0. It follows that $\Phi_T \leq -Y/200$ on the second interval. Thus, whenever

$$m > \eta n, \quad n - m > n/32, \quad (7.24)$$

we have $Y \geq \eta/2$ for all sufficiently large n . Indeed,

$$\frac{m}{n} = \frac{\alpha Y - I}{\alpha - T} \leq \frac{\alpha Y}{\alpha - T},$$

so $m > \eta n$ implies

$$Y > \eta \frac{\alpha - T}{\alpha} \geq \frac{\eta}{2}.$$

The leading term is at most $-k_{\text{co}}\alpha Y/5000$, whereas $\ln(e/Y) = O(\ln \ln n)$, $\ln \ln n/\alpha = o(1)$, and $\ln n = o(k_{\text{co}}\alpha \eta)$. It therefore dominates both error terms in (7.20). Uniformly in this range,

$$D(\ell) \leq e^{-ck_{\text{co}} \ln \ln n}. \quad (7.25)$$

There are only $(k_{\text{co}} + 1)^4$ vectors, so their total is $o(1)$.

Proof: the full corner. The *full corner* is the regime in which the selected common blocks occupy all but a small residual vertex mass, namely $v = n - m \leq n/32$. Uniformly for the four sizes, $\mu_{u_i}(n) \leq n^{6+o(1)}$ by (2.2) and (2.8), while

$$\mu_{u_i}(v) \leq \mu_{u_i}(n)(v/n)^{u_i} \leq n^{-4+o(1)}. \quad (7.26)$$

Here $u_i = (2/\ln 2 + o(1))\ln n$ and $32^{-u_i} = n^{-10+o(1)}$. Consequently $\mu_{u_i}(v + u_i) \leq n^{-4+o(1)}$ at every intermediate step while $v \leq n/32$. Since $k_i \leq n$ and $(h_i + 1)^2 \geq 1$, the right side of (7.6) is at most $2n^{-3+o(1)} < 1$. Thus, in any order of adding residual blocks, $B(h)$ never increases from $B(0) = 1$, and hence $B(h) \leq 1$. Equations (5.19) and (7.5) give

$$D(k - h) \leq e^{-czk_{co}}. \quad (7.27)$$

Summing $(k_{co} + 1)^4$ residual vectors proves an $o(1)$ contribution. Together with (7.11) and (7.25), this proves (7.7). $\square$

8. CANONICAL HIGH CELLS AND DENSE ENDPOINT TRANSPORTATION

CONTRACT FOR THIS SECTION.

The input is the exact overlap sum (6.6), after Section 7 has removed the contribution of common whole classes. The output is a bound for the sum of the weights of all canonical high-cell skeletons, including unequal-type and nonendpoint cells. The local and cycle factors of the unexposed residual configuration are deliberately deferred to Section 9. Keeping those two tasks separate is what prevents the same overlap from being charged both as a high cell and as a residual attachment.

Put

$$u_{\max} = \alpha - 2, \quad R_0 = \lfloor u_{\max}/2 \rfloor. \quad (8.1)$$

In every overlap matrix, the cells of multiplicity greater than R_0 will be called *high cells*. They form a matching: two high cells in one row or column would use more than $u_{\max}$ stubs. The resulting skeleton is canonical: for a given overlap matrix it is exactly the collection of all high cells, together with their multiplicities, and there is no auxiliary choice. Write the resulting canonical high-cell skeleton and its multiplicities as

$$\mathcal{M} = \{a_r b_r : 1 \leq r \leq h\}, \quad r_{a_r b_r} = j_r > R_0, \quad J = \sum_r j_r. \quad (8.2)$$

Selecting the exact stub pairs in these cells has incidence

$$\pi(\mathcal{M}, j) = \frac{\prod_r (s_{a_r})_{j_r} (t_{b_r})_{j_r}}{(n)_J \prod_r j_r!}. \quad (8.3)$$

The corresponding *high-skeleton weight* is

$$w_{\text{hi}}(\mathcal{M}, j) := \pi(\mathcal{M}, j) \prod_{e \in \mathcal{M}} g(j_e). \quad (8.3a)$$

It contains the incidence probability and the signed local rewards of the exposed high cells, but no residual local factor and no cycle-space factor.

Conditional on the exposed pairs, the remaining matching is a uniform configuration model with total degree $n - J$ and the residual row and column degrees. It is restricted to put no further pair in a cell of $\mathcal{M}$ and no cell above R_0 . To see the cancellation explicitly, let

$$s'_a = s_a - \sum_{r: a_r = a} j_r, \quad t'_b = t_b - \sum_{r: b_r = b} j_r,$$

and, for a final overlap matrix r , let r' be obtained by replacing each exposed entry $r_{a_r b_r} = j_r$ by 0. The conditional probability of r' is

$$\frac{\prod_a s'_a! \prod_b t'_b!}{(n-J)! \prod_{a,b} r'_{ab}!}.$$

Multiplying this expression by (8.3), and using $(s_a)_{s_a-s'_a} = s_a!/s'_a!$, $(t_b)_{t_b-t'_b} = t_b!/t'_b!$, and $(n)_J = n!/(n-J)!$, gives

$$\frac{\prod_a s_a! \prod_b t_b!}{n! \prod_{a,b} r_{ab}!},$$

which is exactly (6.2). Thus every overlap matrix occurs exactly once. The canonical choice of all cells above R_0 is what prevents an overlap matrix from being charged to two different skeletons.

We first sum the bare high-skeleton contribution. Index the four sizes as

$$u_i = a - i, \quad 0 \leq i \leq 3, \quad a = \alpha - 2. \quad (8.4)$$

EXACT ENDPOINT BOOKKEEPING.

For a typed full-containment matrix $\mathbf{L} = (\ell_{ij})$, put

$$r_i = \sum_j \ell_{ij}, \quad c_j = \sum_i \ell_{ij}, \quad x_{ij} = \min(u_i, u_j), \quad J(\mathbf{L}) = \sum_{ij} x_{ij} \ell_{ij}. \quad (8.5)$$

Its exact incidence, including its local signed rewards but not residual attachments, is

$$W(\mathbf{L}) = \frac{\prod_i (k_i)_{r_i} \prod_j (k_j)_{c_j}}{\prod_{ij} \ell_{ij}!} \frac{1}{(n)_{J(\mathbf{L})}} \prod_{ij} \left[\frac{(u_i)_{x_{ij}} (u_j)_{x_{ij}}}{x_{ij}!} g(x_{ij}) \right]^{\ell_{ij}}. \quad (8.6)$$

For a margin vector r , define

$$D(r) = \frac{\prod_i (k_i)_{r_i}^2 \prod_i [u_i! g(u_i)]^{r_i}}{\prod_i r_i! (n)_{m_r}}, \quad m_r = \sum_i u_i r_i. \quad (8.7)$$

This is exactly the common-subprofile weight (7.3).

Lemma 8.1 (geometric-mean transportation comparison).

For every feasible $\mathbf{L}$,

$$W(\mathbf{L}) \leq \sqrt{D(r)D(c)} \frac{\sqrt{\prod_i r_i! c_i!}}{\prod_{ij} \ell_{ij}!} \prod_{ij} Q_{ij}^{\ell_{ij}}, \quad (8.8)$$

where $Q_{ii} = 1$ and, if the two sizes are $t = s + d$, $1 \leq d \leq 3$,

$$Q_{ij} = (n+1)^{d/2} \frac{\sqrt{(t)d}}{d!} 2^{-\{ds + \binom{d}{2}\}/2} \leq \frac{\eta_n^d}{d!}, \quad (8.9)$$

with

$$\eta_n = 2^{3/2} \sqrt{\frac{(n+1)a}{2^a}} = O((\ln n)^{3/2}/\sqrt{n}) = o(1). \quad (8.10)$$

Proof. The estimate has two sources. The first is a global falling-factorial comparison: replacing a rectangular overlap table by its two one-sided margin profiles costs only the small off-diagonal factor in (8.12). The second is a local comparison of unequal class sizes: a cell joining sizes that differ by d carries the factor Q_{ij} , and Q_{ij} is small because the random graph penalty $2^{-ds/2}$ dominates the polynomial choices of the extra d vertices.

For integer x , let $f(x) = \ln(n)_x$ and interpolate linearly between successive integers. Its successive slopes are $\ln(n-x)$, which decrease; therefore f is concave, and every slope is at most $\ln(n+1)$. Moreover,

$$\frac{m_r + m_c}{2} = J(\mathbf{L}) + \frac{1}{2} \sum_{ij} |i - j| \ell_{ij}. \quad (8.11)$$

Concavity first gives

$$\frac{f(m_r) + f(m_c)}{2} \leq f\left(\frac{m_r + m_c}{2}\right).$$

Using (8.11) and then moving from $J(\mathbf{L})$ along a distance $\frac{1}{2} \sum_{ij} |i - j| \ell_{ij}$, with slope at most $\ln(n+1)$, gives

$$\frac{f(m_r) + f(m_c)}{2} - f(J(\mathbf{L})) \leq \frac{\ln(n+1)}{2} \sum_{ij} |i - j| \ell_{ij}.$$

Exponentiating this inequality gives

$$\frac{\sqrt{(n)_{m_r}(n)_{m_c}}}{(n)_{J(\mathbf{L})}} \leq (n+1)^{\frac{1}{2} \sum_{ij} |i - j| \ell_{ij}}. \quad (8.12)$$

For $t = s + d$, put $b_u = u!g(u)$. A direct calculation gives

$$\frac{(t)_s (s)_s}{s!} g(s) = b_s \binom{t}{d}, \quad \frac{b_t}{b_s} = (t)_d 2^{ds + \binom{d}{2}}. \quad (8.13)$$

Thus the local factor divided by $\sqrt{b_s b_t}$ is

$$\frac{\sqrt{(t)_d}}{d!} 2^{-\{ds + \binom{d}{2}\}/2}. \quad (8.14)$$

Dividing (8.6) by $\sqrt{D(r)D(c)}$, then using (8.12) and (8.14), proves (8.8)–(8.9). Finally $t \leq a$, $s \geq a - 3$, and

$$2^a = \Theta(n^2 / (\ln n)^2) \quad (8.15)$$

uniformly in the phase. This proves (8.10). $\square$

Lemma 8.2 (sum of all endpoint tables).

Let $\mathcal{P}$ be the finite set of feasible one-sided margin vectors. The first sum below is over all typed full-containment tables $\mathbf{L}$ whose row and column margins lie in $\mathcal{P}$. Then

$$\begin{aligned} \sum_{\mathbf{L}} W(\mathbf{L}) &\leq \exp\{O(\eta_n k_{\text{co}})\} \left(\sum_{\mathbf{r} \in \mathcal{P}} \sqrt{D(\mathbf{r})} \right)^2 \\ &\leq \exp\{O(\sqrt{n \ln n}) + O(\ln n)\} \sum_{\mathbf{r} \in \mathcal{P}} D(\mathbf{r}) \\ &= \exp\{O(\sqrt{n \ln n}) + O(\ln n)\}. \end{aligned} \quad (8.16)$$

Proof. This is the point at which the geometric-mean comparison becomes a genuine sum over all endpoint tables. For fixed margins we forget one set of margin constraints at a time. That enlargement is harmless because all summands are nonnegative, and it converts the constrained table sum into a multinomial sum. Cauchy's inequality then restores symmetry between the two margins.

For fixed margins, write

$$A_{\mathbf{L}} = \frac{\prod_i r_i!}{\prod_{ij} \ell_{ij}!}, \quad C_{\mathbf{L}} = \frac{\prod_j c_j!}{\prod_{ij} \ell_{ij}!}. \quad (8.17)$$

Write $Q^{\mathbf{L}} := \prod_{ij} Q_{ij}^{\ell_{ij}}$. Cauchy's inequality gives

$$\sum_{\mathbf{L}} \sqrt{A_{\mathbf{L}} C_{\mathbf{L}}} Q^{\mathbf{L}} \leq \left(\sum_{\mathbf{L}} A_{\mathbf{L}} Q^{\mathbf{L}} \right)^{1/2} \left(\sum_{\mathbf{L}} C_{\mathbf{L}} Q^{\mathbf{L}} \right)^{1/2}. \quad (8.18)$$

All summands in (8.18) are nonnegative. In the first sum, remove the column-margin constraints while retaining the fixed row margins; this enlarges the summation domain and therefore gives an upper bound. The unrestricted allocation of the r_i row objects among the column types is then counted exactly by the multinomial theorem. The second inequality is obtained symmetrically by removing the row-margin constraints while retaining the column margins. Thus

$$\sum_{\mathbf{L}} A_{\mathbf{L}} Q^{\mathbf{L}} \leq \prod_i \left(\sum_j Q_{ij} \right)^{r_i}, \quad \sum_{\mathbf{L}} C_{\mathbf{L}} Q^{\mathbf{L}} \leq \prod_j \left(\sum_i Q_{ij} \right)^{c_j}. \quad (8.19)$$

Every row and column sum of Q is at most $1 + C\eta_n$, and the total margin is at most k_{co} . Hence the fixed-margin cost is $\exp(O(\eta_n k_{\text{co}}))$. There are only $O(k_{\text{co}}^4)$ margin vectors, and

$$\left(\sum_{\mathbf{r} \in \mathcal{P}} \sqrt{D(\mathbf{r})} \right)^2 \leq O(k_{\text{co}}^4) \sum_{\mathbf{r} \in \mathcal{P}} D(\mathbf{r}). \quad (8.20)$$

Use Lemma 7.1 and $\eta_n k_{\text{co}} = O(\sqrt{n \ln n})$ to finish. $\square$

Lemma 8.3 (all nonendpoint high multiplicities).

After all feasible high-cell multiplicities $R_0 < j \leq \min(u_i, u_j)$ are included,

$$\sum_{(\mathcal{M}, j) \text{ feasible}} w_{\text{hi}}(\mathcal{M}, j) \leq \exp \left\{ o \left(\frac{n}{(\ln n)^4} \right) \right\}.$$

Proof. Replacing a high cell by full containment creates two qualitatively different errors. A small deficit from the endpoint factorizes over the matching and is summable cell by cell; a middle multiplicity must instead be estimated inside the residual configuration model. The proof treats these mechanisms separately and then recombines them through the exact canonical decomposition.

It may help to view the argument as a charging scheme. Every high cell is charged first to the nearest endpoint table. If the cell is close to full containment, the charge is paid immediately by the rapidly decreasing one-cell series (8.25). If it is not close to full containment, it is left in the residual model and paid for by the joint threshold estimate (8.26a), or by the crude small-residual bound (8.29a). The no-double-counting point is that the joint probability estimate is applied before the residual restrictions are relaxed. The proof is organized into five explicit steps: the canonical near–middle split, the summation of near cells, the two residual-mass branches for middle cells, and the final assembly.

STEP I: THE CANONICAL NEAR–MIDDLE SPLIT.

We first record why the following two ranges exhaust the canonical high-skeleton sum. If a high cell joins slots of sizes m and $m + d$, where $0 \leq d \leq 3$, write its multiplicity uniquely as $j = m - e$. Call the cell *near* when $e < m/4$, and *middle* otherwise. Because the high cells form a matching, replacing every near cell by the endpoint multiplicity m produces a feasible typed full-containment table $\mathbf{L}$; the original near cells are recovered from $\mathbf{L}$ and their uniquely determined deficits e . The remaining high cells satisfy

$$R_0 < j = m - e \leq 3m/4 \leq 3a/4.$$

Thus every canonical high skeleton has a unique near part recorded by an endpoint table and a residual middle part. In the upper bounds below we may admit residual choices that do not themselves form a matching or satisfy the no-near-cell restriction, but only after the relevant joint probability has been bounded. Since all summands are nonnegative, these relaxations can only increase the sum.

STEP II: SUMMING NEAR-CONTAINMENT CELLS.

Let the smaller slot size be m , the larger $m + d$, with $0 \leq d \leq 3$. Replacing the endpoint multiplicity m by $m - e$ has exact local ratio

$$R_{m,d}(e) = \frac{\binom{m}{e}}{(d+1) \cdots (d+e)} 2^{-em+e(e+1)/2}. \quad (8.21)$$

For many decorated cells, with total deficit Q , the ratio of the one global denominators is

$$\frac{\binom{n}{J_0}}{\binom{n}{J_0-Q}} = (n - J_0 + Q)_Q \leq n^Q. \quad (8.22)$$

Put

$$A_e := n^e R_{m,d}(e), \quad 0 \leq e < m/4.$$

For $0 \leq e < m/4$, its consecutive ratio is

$$\frac{n(m-e)2^{-m+e+1}}{(e+1)(d+e+1)}. \quad (8.23)$$

The ratio of (8.23) at $e+1$ to its value at e is

$$2 \frac{m-e-1}{m-e} \frac{e+1}{e+2} \frac{d+e+1}{d+e+2}, \quad (8.24)$$

which increases for $e \leq m/4$ and $d \leq 3$. Here is a direct check of the one-turn assertion. If $r(e)$ denotes (8.23), interpolated for real e , then

$$\frac{d^2}{de^2} \ln r(e) = -\frac{1}{(m-e)^2} + \frac{1}{(e+1)^2} + \frac{1}{(d+e+1)^2} > 0 \quad (8.24a)$$

for $0 \leq e \leq m/4$ and all sufficiently large m . Thus the consecutive ratios are log-convex and their maximum is at an endpoint. At $e = 0$ the ratio is $O((\ln n)^3/n)$, by (8.15), and at $e = \lfloor m/4 \rfloor$ it is $n^{-1/2+o(1)}$. The series therefore decreases geometrically and

$$\sum_{1 \leq e < m/4} n^e R_{m,d}(e) = O((\ln n)^3/n). \quad (8.25)$$

FROM ONE NEAR CELL TO A FULL SKELETON.

The passage from this per-cell estimate to the full skeleton sum is as follows. For a full-containment table $\mathbf{L}$, temporarily distinguish its endpoint cells, write $\text{cells}(\mathbf{L})$ for the resulting multiset, and let $\mathcal{S}(\mathbf{L})$ be all near-containment skeletons obtained by assigning to each such cell either $e = 0$ or $1 \leq e < m/4$. If $w(S)$ is its bare incidence and local weight, (8.21)–(8.25) give the nonnegative bound

$$\sum_{S \in \mathcal{S}(\mathbf{L})} w(S) \leq W(\mathbf{L}) \prod_{c \in \text{cells}(\mathbf{L})} \left(1 + \sum_{1 \leq e < m_c/4} n^e R_{m_c, d_c}(e) \right). \quad (8.25a)$$

Thus a near-containment decoration is simply an endpoint cell of smaller size m whose multiplicity has been lowered to $m - e$ by fewer than $m/4$ stubs; the terminology records its geometric proximity to full containment.

Distinguishing and then forgetting identical typed cells is exactly the multinomial expansion, so there is no additional multiplicity. Since a high skeleton is a matching and has at most k_{co} cells, (8.25a) is at most $W(\mathbf{L})$ times

$$(1 + O((\ln n)^3/n))^{k_{\text{co}}} = \exp(O((\ln n)^2)). \quad (8.26)$$

STEP III: MIDDLE CELLS WITH LARGE RESIDUAL MASS.

It remains to cover the middle strip, rather than silently importing an equitable-profile estimate. First expose a near-containment skeleton S . Its underlying set of high cells is a matching; write $\mathcal{M}(S)$ for that matching after multiplicities are forgotten, and let m_0 be the residual stub mass. Let $\mathcal{N}(S)$ be the residual event that there is no further near-containment cell. On $\mathcal{N}(S)$, every unexposed high cell is in the range $R_0 < j \leq 3a/4 + O(1)$. Define $E_{\text{mid}}(S)$ as the residual expectation of the product of the local factors truncated to this middle range, multiplied by $\mathbf{1}_{\mathcal{N}(S)}$. If $\tilde{g}$ denotes this truncated factor, then on $\mathcal{N}(S)$

$$\tilde{g}(r_{ab}) \leq 1 + \sum_{R_0 < j \leq 3a/4 + O(1)} g(j) \mathbf{1}_{\{r_{ab} \geq j\}}.$$

Expanding the product produces nonnegative monomials, each of which chooses distinct residual cells and at most one threshold in each selected cell. Apply Lemma 6.2 jointly to the entire threshold family before changing any residual restriction. Only afterwards do we discard $\mathbf{1}_{\mathcal{N}(S)}$: this can only increase each nonnegative monomial. The local factor remains truncated to the stated middle range, so the relaxation does not create a new charge for a newly admitted near-containment cell. If $m_0 \geq n/(\ln n)^6$, expand over

distinct residual cells and their threshold demands. With $\theta_{ab} = e d_a d'_b / m_0 \leq e a^2 / m_0$, Lemma 6.2 is applied jointly before the constraints are dropped, and gives

$$\begin{aligned} E_{\text{mid}}(S) &\leq \sum_{\substack{D \text{ distinct residual cells} \\ R_0 < j_{ab} \leq 3a/4 + O(1)}} \prod_{ab \in D} g(j_{ab}) \mathbb{P}(r_{ab} \geq j_{ab} \text{ for every } ab \in D) \\ &\leq \prod_{ab} \left(1 + \sum_{R_0 < j \leq 3a/4 + O(1)} g(j) \frac{\theta_{ab}^j}{j!} \right) \leq \exp(\Xi_4), \end{aligned} \quad (8.26a)$$

where

$$\Xi_4 \leq k_{\text{co}}^2 \sum_{a/2 < j \leq 3a/4 + O(1)} g(j) \frac{(e a^2 / m_0)^j}{j!}. \quad (8.27)$$

Put $\ell_n = \log_2 n$ and $j = x \ell_n$. Uniformly for $1 + o(1) \leq x \leq 3/2 + o(1)$,

$$\begin{aligned} \log_2 \left[k_{\text{co}}^2 g(j) \frac{(e a^2 / m_0)^j}{j!} \right] &\leq 2\ell_n + \frac{j^2}{2} - j \log_2 m_0 + O(j \log_2 a) \\ &\leq -c \ell_n^2. \end{aligned} \quad (8.28)$$

Indeed, the quadratic coefficient is $x^2/2 - x \leq -3/8 + o(1)$ on this interval. Hence $\Xi_4 = 2^{-\Omega(\ell_n^2)}$.

STEP IV: MIDDLE CELLS WITH SMALL RESIDUAL MASS.

If instead $m_0 < n/(\ln n)^6$, leave the entire residual matching unexposed. For an upper bound, replace $\mathbf{1}_{\mathcal{N}(S)}$ by 1 and the truncated local product by the full product $\prod_e g(r_e)$; both replacements enlarge the nonnegative integrand. Every residual degree is at most a , so every residual cell has $r_e \leq a$ and therefore $\binom{r_e}{2} \leq ((a-1)/2)r_e$. The near skeleton is a matching, so pointwise

$$\beta(\mathcal{M}(S) \cup H_{\text{res}}) \leq |E(H_{\text{res}})| \leq m_0/2, \quad \sum_e \binom{r_e}{2} \leq (a-1)m_0/2. \quad (8.29)$$

All remaining local and topological factors together are therefore at most $\exp(O(am_0)) = \exp(O(n/(\ln n)^5))$. Since $g \geq 1$ and $\beta \geq 0$, summing the residual matching probability first gives the explicit conditional bound

$$E_{\text{mid}}(S) \leq \mathbb{E}_{\text{res}} \left[\prod_e g(r_e) 2^{\beta(\mathcal{M}(S) \cup H_{\text{res}})} \right] \leq \exp(Cam_0). \quad (8.29a)$$

STEP V: ASSEMBLY.

Combining (8.16), (8.25a)–(8.26a), and the two residual-mass branches now gives the global inequality

$$\sum_{(\mathcal{M}, j) \text{ feasible}} w_{\text{hi}}(\mathcal{M}, j) \leq e^{O((\ln n)^2)} \left(\sum_{\mathbf{L}} W(\mathbf{L}) \right) \max\{e^{\Xi_4}, e^{Cn/(\ln n)^5}\}. \quad (8.29b)$$

Therefore

$$\sum_{(\mathcal{M}, j) \text{ feasible}} w_{\text{hi}}(\mathcal{M}, j) \leq \exp\{O(\sqrt{n \ln n}) + O((\ln n)^2) + O(n/(\ln n)^5)\} = \exp\{o(n/(\ln n)^4)\}. \quad (8.30)$$

This is the skeleton-sum half of the factorization (9.2); no residual attachment has yet been charged. $\square$

9. ALL RESIDUAL LOCAL AND CYCLE ATTACHMENTS

CONTRACT FOR THIS SECTION.

Section 8 supplied the sum of the high-skeleton weights w_{hi} , but deliberately left the unexposed local and cycle-space factors unpaid. Here we condition on one feasible skeleton. The remaining stubs form a finite bipartite configuration model with prescribed residual degrees, and $\mathcal{A}(\mathcal{M}, j)$ records exactly its residual attachment. Lemma 9.1 bounds this quantity uniformly; Proposition 9.2 then multiplies that uniform bound by the skeleton sum (8.30).

EXACT CONDITIONAL DECOMPOSITION.

Fix an actual canonical high skeleton $(\mathcal{M}, j)$ as in (8.2), and let $m_0 = n - J$ be the residual total degree. Its residual degrees are at most $u_{\max}$. Define

$$\mathcal{A}(\mathcal{M}, j) = \mathbb{E}_{\text{res}} \left[\left(\prod_e g(r'_e) \right) 2^{\beta(\mathcal{M} \cup H_{\text{res}})} \mathbf{1}_{\mathcal{E}(\mathcal{M}, j)} \right], \quad (9.1)$$

where $\mathcal{E}(\mathcal{M}, j)$ imposes the residual cap $r'_e \leq R_0$ and forbids a further pair in a cell of $\mathcal{M}$. By (6.4), (8.3), and the exact conditional law, the normalized moment is exactly

$$\frac{\mathbb{E}[(Z_{\mathbf{k}}^{\text{sgn}})^2]}{\mathbb{E}[Z_{\mathbf{k}}^{\text{sgn}}]^2} = \sum_{(\mathcal{M}, j)} w_{\text{hi}}(\mathcal{M}, j) \mathcal{A}(\mathcal{M}, j). \quad (9.2)$$

The architecture is now visible in a single line: Section 8 bounds the sum of w_{hi} , Lemma 9.1 bounds the supremum of $\mathcal{A}$, and Proposition 9.2 combines the two bounds.

The sum in (9.2) is over the finite set of feasible canonical high skeletons: typed matchings $\mathcal{M}$, allowed multiplicities $j_e > R_0$, and the induced nonnegative residual row and column degrees.

Here $\mathbb{E}_{\text{res}}[\cdot]$ denotes expectation only over the unexposed matching of the remaining stubs. A *residual attachment* is this conditional contribution: the local factors $g(r'_e)$ record residual cell multiplicities, while $2^{\beta(\mathcal{M} \cup H_{\text{res}})}$ records the associated cycle-space contribution after the high skeleton has already been fixed.

UNIFORM ASYMPTOTIC BOUND.

We next bound the exact conditional quantity (9.1) uniformly over its finite family of feasible skeletons.

Lemma 9.1 (uniform attachment bound).

There is a deterministic sequence $\varepsilon_n \rightarrow 0$, independent of the canonical high skeleton, such that for every feasible skeleton and all sufficiently large n ,

$$\mathcal{A}(\mathcal{M}, j) \leq \exp \left\{ \varepsilon_n \frac{n}{(\ln n)^4} \right\}. \quad (9.3)$$

More precisely, for an absolute C , it is at most $\exp(C(\ln n)^8)$ when $m_0 \geq n/(\ln n)^6$ and at most $\exp(Cn/(\ln n)^5)$ otherwise. No lower bound on $\mathcal{A}$ is asserted or needed.

Proof. LARGE RESIDUAL MASS.

There are two possible ways a residual cell can matter. It may contribute a large local multiplicity factor $g(r'_e)$, or it may be selected as part of an even residual subgraph contributing to the cycle-space

term. The expansion below assigns these two costs to different weights. The quantities $\lambda_{ab}^{\text{loc}}$ pay for multiplicity increments $r'_e \geq 3$, whereas q_{ab} pays for a residual edge used by the even-subgraph sum.

Suppose $m_0 \geq n/(\ln n)^6$. Since

$$u_{\max} = (2 + o(1)) \log_2 n, \quad \log_2 m_0 = \log_2 n - O(\log_2 \ln n), \quad (9.4)$$

we have $u_{\max} < 3 \log_2 m_0$ for all sufficiently large n . For a residual cell outside $\mathcal{M}$, put

$$\theta_{ab} = \frac{e d_a d'_b}{m_0}, \quad \Delta_x = g(x) - g(x-1), \quad (9.5)$$

and

$$\lambda_{ab}^{\text{loc}} = \sum_{x=3}^{R_0} \Delta_x \frac{\theta_{ab}^x}{x!}, \quad q_{ab} = \frac{\theta_{ab}^2}{2} + \lambda_{ab}^{\text{loc}}. \quad (9.6)$$

Set both quantities to zero on $\mathcal{M}$. Let $\theta_* = e u_{\max}^2 / m_0$. The ratio of consecutive majorants $g(x) \theta_*^x / x!$ is

$$\frac{2^x \theta_*}{x+1}, \quad (9.7)$$

and increases with x . Hence the sequence is log-convex. Its decreasing head is dominated by the $x = 3$ term. Consecutive values of (9.7) grow by the factor $2(x+1)/(x+2) \geq 8/5$, so only $O(1)$ transition terms can have ratio between $1/2$ and 1 . Its increasing tail is dominated by its right endpoint, and

$$\log_2 \left(g(R_0) \frac{\theta_*^{R_0}}{R_0!} \right) \leq \frac{R_0^2}{2} - R_0 \log_2 m_0 + O(R_0 \log_2 u_{\max}) = -\Omega(R_0 \log n), \quad (9.8)$$

using $R_0 = u_{\max}/2$ and (9.4). Thus, uniformly in the cell,

$$\lambda_{ab}^{\text{loc}} \leq C \theta_{ab}^3, \quad q_{ab} \leq C \theta_{ab}^2. \quad (9.9)$$

Expand each capped local factor as

$$g(r'_e) = 1 + \sum_{x=3}^{R_0} \Delta_x \mathbf{1}_{\{r'_e \geq x\}}. \quad (9.10)$$

If an edge is selected in a fixed even subgraph, use instead

$$\mathbf{1}_{\{r'_e \geq 2\}} g(r'_e) = \mathbf{1}_{\{r'_e \geq 2\}} + \sum_{x=3}^{R_0} \Delta_x \mathbf{1}_{\{r'_e \geq x\}}. \quad (9.11)$$

In the product expansion, a selected cycle-space edge contributes either the base demand $\mathbf{1}_{\{r'_e \geq 2\}}$ or one increment term $\Delta_x \mathbf{1}_{\{r'_e \geq x\}}$, but not both in the same monomial. Thus a cell of multiplicity at least three is not charged simultaneously as a separate double-cell demand and as a higher-threshold local-factor demand. Let R be the finite potential residual support graph consisting of all residual row–column cells, and let $C_{\text{even}}(\mathcal{M}, R)$ be the finite family of edge sets $F \subseteq E(\mathcal{M} \cup R)$ having even degree at every vertex. The edges of $\mathcal{M}$ have weight one below.

For clarity, the derivation of the next inequality has four steps. First, expand 2^β by the cycle-space identity (6.7). Second, use (9.10) on edges outside the chosen even set and (9.11) on its selected residual edges. Third, apply Lemma 6.2 jointly to every resulting threshold demand before removing the cap or the no-return restriction. Fourth, factor the remaining local increments cell by cell. Every coefficient and event indicator is nonnegative, so each relaxation preserves the upper-bound direction. This gives

$$\mathcal{A}(\mathcal{M}, j) \leq e^{\Lambda_{\text{loc}}} \sum_{F \in \mathcal{C}_{\text{even}}(\mathcal{M}, R)} \prod_{e \in F \setminus \mathcal{M}} q_e, \quad \Lambda_{\text{loc}} = \sum_e \lambda_e^{\text{loc}}. \quad (9.12)$$

The right side now has a clean division of labour: the exponential term pays for higher local multiplicities, while the weighted even-subgraph sum pays for the cycle space. The remaining proof bounds these two quantities independently.

By (9.9),

$$\sum_{a,b} \theta_{ab}^3 = \frac{e^3}{m_0^3} \left(\sum_a d_a^3 \right) \left(\sum_b (d'_b)^3 \right).$$

Every residual degree is at most u_{max} , and both degree sums equal m_0 . Therefore $\sum_a d_a^3 \leq u_{\text{max}}^2 m_0$ and $\sum_b (d'_b)^3 \leq u_{\text{max}}^2 m_0$, which gives

$$\Lambda_{\text{loc}} \leq \frac{C}{m_0^3} \left(\sum_a d_a^3 \right) \left(\sum_b (d'_b)^3 \right) \leq C \frac{u_{\text{max}}^4}{m_0}. \quad (9.13)$$

For a fixed row a , the quadratic bound in (9.9) also gives

$$\sum_b \theta_{ab}^2 = \frac{e^2 d_a^2}{m_0^2} \sum_b (d'_b)^2 \leq C \frac{u_{\text{max}}^3}{m_0},$$

and the same calculation holds for every column. Hence

$$\max_a \sum_b q_{ab} \vee \max_b \sum_a q_{ab} \leq C \frac{u_{\text{max}}^3}{m_0} =: \tau = o(1). \quad (9.14)$$

CYCLE-KERNEL ESTIMATE.

Every even edge set has a deterministic decomposition into edge-disjoint simple cycles, and the map from an even edge set to that chosen cycle family is injective because the original edge set is recovered by taking the union. We now enlarge the image of this map to all finite families of simple cycles, allowing cycles in the family to share edges. Since all cycle weights are nonnegative, this relaxation can only increase the sum. With $w(C) = \prod_{e \in C \setminus \mathcal{M}} q_e$ and $w(F) = \prod_{e \in F \setminus \mathcal{M}} q_e$, summing over all cycle families and using $1 + x \leq e^x$ gives

$$\sum_{F \text{ even}} \prod_{e \in F \setminus \mathcal{M}} q_e \leq \exp \left\{ \sum_{C \text{ simple cycle}} \prod_{e \in C \setminus \mathcal{M}} q_e \right\}. \quad (9.15)$$

Let K be the symmetric nonnegative kernel on the disjoint union of the residual row and column vertices, with $K(a, b) = K(b, a) = q_{ab}$ on residual cells and zero elsewhere. Equation (9.14) says that its maximum row sum is at most τ .

For cycles disjoint from $\mathcal{M}$, orient the cycle and distinguish one row vertex. A cycle of length $2r$, $r \geq 2$, is thereby encoded by one of the alternating K -walks of length $2r$ from that vertex back to itself. Dropping simplicity and the final return constraint only enlarges the family. There are at most n possible distinguished row vertices, and successive row-sum bounds give $n \sum_{r \geq 2} \tau^{2r}$. Hence

$$\sum_{C: C \cap \mathcal{M} = \emptyset} \prod_{e \in C} q_e \leq \frac{n \tau^4}{1 - \tau^2}. \quad (9.16)$$

For the mixed cycles, define the positive residual-walk kernel

$$S = \sum_{\ell \geq 1} K^\ell, \quad \|S\|_\infty \leq b := \sum_{\ell \geq 1} \tau^\ell = \frac{\tau}{1-\tau}. \quad (9.17)$$

This series is finite entrywise for all sufficiently large n , since $\tau = o(1)$, and its entries are the total weights of positive residual walks with prescribed endpoints. Let P be the symmetric adjacency kernel of the matching $\mathcal{M}$. It is a partial permutation, so $\|P\|_\infty \leq 1$.

Bound the mixed-cycle sum by first marking and orienting one of its matching edges; this only overcounts. Across all cycles there are at most $2h$ possible oriented starts. If a marked cycle uses exactly $r \geq 1$ matching edges, cutting it at those edges gives r nonempty residual path segments: consecutive matching edges cannot occur because $\mathcal{M}$ is a matching. After the distinguished edge, the ordered segment data injectively records the marked cycle. If simplicity, distinctness of the later matching edges, and the final endpoint constraint are discarded, the total remaining weight for the fixed oriented start is at most

$$\|S(PS)^{r-1}\|_\infty \leq b^r.$$

The initial choice is the only factor depending on h ; every later matching traversal is already represented by P . Summing over the oriented initial edge and over r gives

$$\sum_{C: C \cap \mathcal{M} \neq \emptyset} \prod_{e \in C \setminus \mathcal{M}} q_e \leq 2h \sum_{r \geq 1} b^r \leq Ch\tau. \quad (9.18)$$

Every high cell uses more than $u_{\max}/2$ stubs, so $h < 2n/u_{\max}$. At the cutoff $m_0 \geq n/(\ln n)^6$,

$$\tau = O((\ln n)^9/n), \quad h\tau = O((\ln n)^8), \quad \frac{u_{\max}^4}{m_0} + n\tau^4 = o(1).$$

Combining (9.12)–(9.18),

$$\mathcal{A}(\mathcal{M}, j) \leq \exp \left[C \left\{ \frac{u_{\max}^4}{m_0} + n\tau^4 + h\tau \right\} \right] \leq \exp(C'(\ln n)^8) \quad (9.19)$$

at the cutoff $m_0 \geq n/(\ln n)^6$.

SMALL RESIDUAL MASS.

Suppose $m_0 < n/(\ln n)^6$. Since $\mathcal{M}$ is a matching, it is a forest, and adding a residual support edge increases cycle rank by at most one. Hence

$$\beta(\mathcal{M} \cup H_{\text{res}}) \leq |E(H_{\text{res}})| \leq m_0/2. \quad (9.20)$$

Also, pointwise,

$$\sum_e \binom{r'_e}{2} \leq \frac{u_{\max} - 1}{2} \sum_e r'_e = \frac{u_{\max} - 1}{2} m_0. \quad (9.21)$$

Equations (6.5), (9.20), and (9.21) bound the integrand in (9.1) by $2^{u_{\max} m_0/2}$. Therefore

$$\mathcal{A}(\mathcal{M}, j) \leq \exp(O(u_{\max} m_0)) \leq \exp(Cn/(\ln n)^5). \quad (9.22)$$

Both exponents in (9.19) and (9.22) are $o(n/(\ln n)^4)$. $\square$

Proposition 9.2 (normalized signed second moment).

For the exact midpoint profile,

$$\Lambda_n := \ln \frac{\mathbb{E}[(Z_{\mathbf{k}}^{\text{sgn}})^2]}{\mathbb{E}[Z_{\mathbf{k}}^{\text{sgn}}]^2} = o(n/(\ln n)^4). \quad (9.23)$$

Proof. Equation (9.2) is an exact canonical decomposition. Lemma 8.3 sums all bare high skeletons by $\exp(o(n/(\ln n)^4))$, including the empty skeleton, all dense endpoint tables, every near-containment decoration, and the two middle-strip regimes. Lemma 9.1 bounds the conditional residual factor by the same exponential scale uniformly over every skeleton. Thus (9.2) gives the explicit estimate

$$\frac{\mathbb{E}[(Z_{\mathbf{k}}^{\text{sgn}})^2]}{\mathbb{E}[Z_{\mathbf{k}}^{\text{sgn}}]^2} \leq \left(\sup_{(\mathcal{M}, j)} \mathcal{A}(\mathcal{M}, j) \right) \sum_{(\mathcal{M}, j)} w_{\text{hi}}(\mathcal{M}, j) = \exp\{o(n/(\ln n)^4)\}.$$

This proves the upper bound in (9.23). The normalized second moment is at least one by variance, so its logarithm is nonnegative and (9.23) follows. $\square$

10. RARE-EVENT AMPLIFICATION

Proposition 9.2 and (1.4) give the seed

$$\mathbb{P}(Z_{\mathbf{k}}^{\text{sgn}} > 0) \geq e^{-\Lambda_n}. \quad (10.1)$$

This is only a possibly rare seed. The purpose of this section is to turn it into a high-probability cocoloring event while adding fewer parts than the root separation can absorb.

The proof of Heckel (2025, Theorem 1) also begins with a rare second-moment seed and then uses vertex-exposure concentration of $\zeta(G_n)$ to obtain a high-probability statement. We use the same seed-to-typical strategic principle, but not that theorem as a black box: Lemma 10.2 proves the quantitative implication needed here for an arbitrary seed exponent Λ_n , and Lemma 10.1 supplies the simultaneous leftover coloring that controls the added parts.

The ordinary-coloring concentration argument motivating this amplification appears in Scott (2017, Theorem 1). Lemmas 10.1 and 10.2 prove the precise simultaneous-leftover and rare-seed forms required here.

By the observation after (5.19), this implies

$$\mathbb{P}(\zeta(G_n) \leq k_{\text{co}}) \geq e^{-\Lambda_n}. \quad (10.2)$$

We now prove the amplification needed to turn this possibly rare event into a typical one.

Lemma 10.1 (simultaneous leftover coloring).

There is an absolute C such that, with probability $1 - o(1)$, every $S \subseteq [n]$ satisfies

$$\chi(G_n[S]) \leq C \frac{|S|}{\ln n} + n^{1/3}. \quad (10.3)$$

Proof. Let H be the complement of G_n . Put $u_0 = \lceil n^{1/4} \rceil$. For any fixed u_0 -set, (1.5) gives probability $\exp(-\Omega(u_0^2))$ that its H -edge density is below $1/4$. There are at most $\binom{n}{u_0}$ such sets, and for an absolute $c > 0$,

$$\binom{n}{u_0} e^{-cu_0^2} \leq \exp\{u_0 \ln(en/u_0) - cu_0^2\} = o(1).$$

Thus a union bound shows that, with probability $1 - o(1)$, every u_0 -set has density at least $1/4$. By averaging over its u_0 -subsets, the same is then true of every larger set.

Start with any vertex set of size at least $n^{1/3}$. While its current size is at least u_0 , choose a vertex of maximum H -degree and replace the current set by its H -neighborhood there. Density at least $1/4$ implies that this neighborhood has size at least one quarter of the current size, up to an additive one. If s_t is the current size after t choices, then

$$s_{t+1} \geq (s_t - 1)/4, \quad s_t \geq 4^{-t} s_0 - 1/3. \quad (10.3a)$$

The chosen vertices form a clique in H : after a vertex is chosen, the current set is replaced by its H -neighbourhood, so every later choice is adjacent in H to every earlier one. Starting from any $\lceil n^{1/3} \rceil$ -subset, (10.3a) stays above $n^{1/4}$ for at least $\lfloor \ln n / (13 \ln 4) \rfloor$ steps. Thus the procedure constructs at least $c \ln n$ vertices for an absolute $c > 0$, and these vertices form an independent set in G_n .

For an arbitrary S , repeatedly remove such independent sets and give each a new color until fewer than $n^{1/3}$ vertices remain; color the rest singly. Every removed set has at least $c \ln n$ vertices, so the total number of colors is at most

$$\left\lceil \frac{|S|}{c \ln n} \right\rceil + n^{1/3}.$$

After enlarging the absolute constant, this is (10.3), simultaneously for every S . $\square$

Lemma 10.2 (amplification from a seed).

Suppose deterministic $k_n, \Lambda_n \geq 0$ satisfy

$$\mathbb{P}(\zeta(G_n) \leq k_n) \geq e^{-\Lambda_n}. \quad (10.4)$$

There is a deterministic $\varepsilon_n = o(1)$, independent of k_n, Λ_n, r , such that, uniformly for every deterministic choice $r = r(n) > 0$,

$$\begin{aligned} \mathbb{P}\left(\zeta(G_n) > k_n + C\left(\frac{\sqrt{n\Lambda_n} + \sqrt{nr}}{\ln n} + n^{1/3} + 1\right)\right) \\ \leq e^{-r} + \varepsilon_n. \end{aligned} \quad (10.5)$$

Proof. For an induced set W , let $\zeta(W) = \zeta(G_n[W])$, and define

$$S_k = \max\{|W| : \zeta(W) \leq k_n\}. \quad (10.6)$$

Expose the random graph in $n - 1$ independent vertex blocks, where block v contains the edges from v to earlier vertices. Changing one block changes S_k by at most one. Indeed, after deleting the affected vertex, every feasible induced set loses at most one vertex, and the two graph configurations agree on all remaining edges. Thus a maximizer in either configuration yields a feasible set of size at least one less in the other configuration. Therefore (1.3) applies.

Since $S_k = n$ exactly when $\zeta(G_n) \leq k_n$, (10.4) and the upper one-sided bounded-differences tail give

$$e^{-\Lambda_n} \leq \mathbb{P}(S_k - \mathbb{E}[S_k] \geq n - \mathbb{E}[S_k]) \leq \exp\left\{-\frac{2(n - \mathbb{E}[S_k])^2}{n-1}\right\}.$$

Taking logarithms and rearranging imply

$$n - \mathbb{E}[S_k] \leq \sqrt{(n-1)\Lambda_n/2}. \quad (10.7)$$

The lower tail with radius $\sqrt{(n-1)r/2}$ gives, except with probability e^{-r} ,

$$n - S_k \leq \sqrt{(n-1)\Lambda_n/2} + \sqrt{(n-1)r/2}. \quad (10.8)$$

Choose a maximizing set W and put $V_{\text{left}} = [n] \setminus W$. Concatenating a k_n -part cocoloring of $G[W]$ with an ordinary coloring of $G[V_{\text{left}}]$ gives

$$\zeta(G_n) \leq k_n + \chi(G_n[V_{\text{left}}]). \quad (10.9)$$

Intersect (10.8) with the simultaneous event in Lemma 10.1 and apply (10.3). Since $|V_{\text{left}}| = n - S_k$, equations (10.3), (10.8), and (10.9) give exactly the number of additional parts displayed in (10.5). The simultaneous event's failure probability is the same deterministic $\varepsilon_n = o(1)$ for every choice of the three deterministic parameters. This proves (10.5). $\square$

Apply Lemma 10.2 to (10.2). Put

$$B_n = \frac{n}{(\ln n)^4}, \quad r_n = \sqrt{B_n}. \quad (10.10)$$

Then $r_n \rightarrow \infty$, and Proposition 9.2 gives

$$\frac{\sqrt{n\Lambda_n}}{\ln n} = o(n/(\ln n)^3), \quad \frac{\sqrt{nr_n}}{\ln n} = o(n/(\ln n)^3), \quad n^{1/3} = o(n/(\ln n)^3). \quad (10.11)$$

Define the deterministic sequence

$$a_n = C \left(\frac{\sqrt{n\Lambda_n} + \sqrt{nr_n}}{\ln n} + n^{1/3} + 1 \right). \quad (10.12)$$

Then (10.11) and Lemma 10.2 give

$$a_n = o(n/(\ln n)^3), \quad \mathbb{P}(\zeta(G_n) > k_{\text{co}} + a_n) \leq e^{-r_n} + \varepsilon_n \rightarrow 0. \quad (10.13)$$

11. COMPLETION OF THE PROOF

Intersect $\{\chi(G_n) > k_{\chi}^-\}$, whose probability tends to one by (4.5), with $\{\zeta(G_n) \leq k_{\text{co}} + a_n\}$, whose probability tends to one by (10.13). No independence is required; a union bound suffices. Equation (5.20) then gives, with high probability,

$$\chi(G_n) - \zeta(G_n) \geq \left(\frac{(\ln 2)^2 \gamma_4}{16} - o(1) \right) \frac{n}{(\ln n)^3}. \quad (11.1)$$

For every sufficiently large n , the right side is at least

$$c_* \frac{n}{(\ln n)^3}, \quad c_* := \frac{(\ln 2)^2 \gamma_4}{32} = \frac{(\ln 2)^2}{32} \ln \frac{200}{153} > 0. \quad (11.2)$$

This proves (0.1). Since $n/(\ln n)^3$ tends to infinity, for every fixed K ,

$$\mathbb{P}(\chi(G_n) - \zeta(G_n) \geq K) \rightarrow 1. \quad (11.3)$$

Every estimate used above is uniform for $\delta \in [0, 1)$ and for the exact tangent-rounded finite- n profile. The conclusion therefore holds along all sufficiently large integers, including sequences approaching either side of every independence-number jump. The phase has not been averaged away or discarded: the result is a full-sequence high-probability statement, not a density-subset or subsequence statement. ■

REPRODUCIBILITY AND FORMALISATION

The complete proof sources and reproducibility material are collected in the numbered dossiers of the public Erdos repository; the present paper is in 625/. All theorem statements and proof text are available with the manuscript sources. The dossier includes the TeX source, bibliography, reproducible build material, audit records, and the current partial Lean 4 formalisation.

The 625/verification/ directory additionally contains an independently written Python checker, its reproduced output, and a Markdown verification report. These files provide supplementary finite consistency checks and an auditable record of what was tested; they are not a substitute for external peer review or for a complete formal proof.

The Lean checkpoint, developed using ChatGPT-5.6 and Aristotle (Achim et al., 2025), is provided as a self-contained source file. Each formalised statement is decomposed into smaller claims, with its hypotheses and conclusion checked explicitly. This is a partial machine-checked formalisation, not a claim that the entire paper has been formalised.

AI ASSISTANCE

The proof and formalisation were completed with extensive assistance from ChatGPT-5.6 Sol Ultra and Aristotle. Separate adversarial verification passes were run with ChatGPT-5.6 Sol Pro; these passes are not external peer review. In particular, ChatGPT-5.6 proposed the four-size signed profile with class sizes $\alpha - 2, \alpha - 3, \alpha - 4, \alpha - 5$, which led to the phase-uniform first-moment separation used in Section 5. The author reviewed every incorporated suggestion and assumes full responsibility for the arguments, citations, and final manuscript.

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The author declares no competing interests.

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